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EVOLUTION OF SCALAR FIELD COSMOLOGICAL PERTURBATIONS

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ABSTRACT

A method of dealing with cosmological perturbations was suggested recently which allows a simple way of choosing the gauge appropriate to each separate problem. Using this method, if we have one solution in a particular gauge we can easily derive the rest of the solutions in all the other gauges. In this paper we apply the method to a medium dominated by *a minimally coupled scalar field*. By considering full metric perturbations we are doing scalar field quantum field theory in a spatially flat homogeneous and isotropic background spacetime, now including the metric perturbations as additional quantum fields consistently excited from the scalar field perturbations. Asymptotic solutions in a general potential case, and the exact solutions in an exponential potential case—thus a power-law expanding background case—are presented in tabular forms. Perturbations in de Sitter space—thus an exact exponentially expanding background case—need a separate treatment which is also presented. It is interesting to note that it is in the case that takes into account the full metric perturbations which allows more self-consistent analysis, compared to the one based on pure background metric. We identify the uniform-curvature gauge as providing the simple analysis in the scalar field perturbations. The full solutions may provide a wider perspective on the subject.

Subject headings: cosmology: theory — early universe — large-scale structure of universe

1. INTRODUCTION

The original analysis of cosmological perturbations in Lifshitz (1946) was based on choosing a particular gauge condition called a synchronous gauge. This gauge is most widely used in the literature despite its immediate drawback that the gauge condition does not completely remove the gauge mode. In many cases this remaining gauge mode caused confusion in the literature. Of course, since one can remove the remaining unphysical gauge modes by tracing the behavior of those modes, by itself it is not a serious problem. There exist many other convenient gauge choices introduced in the literature. Although the synchronous gauge is the most popular one, curiously it is the only (out of many convenient choices) gauge choice where the gauge mode was not automatically removed. Although in many papers the existence of the gauge freedom was regarded as troublesome, being a gauge theory, the existence of the gauge freedom can be used as an advantage in dealing with each particular problem. For example, some gauges are appropriate (sometimes even superior) for tackling some particular problems. This point will become clear in this paper, where we investigate scalar field perturbations in various gauge choices; we will find out that a particular gauge allows the most simple analysis.

To use this gauge freedom as an advantage in this way, it is desirable to write the equations without choosing any gauge. This procedure also allows us to translate a set of solutions (known in one gauge) to all the other gauges in a simple manner. In this way the remaining gauge modes in the synchronous gauge will be automatically traced. The gauge-invariant method introduced in Bardeen (1980) is, in a sense, a step toward this (we may call) *gauge nonspecific method*. Indeed, this gauge nonspecific method was introduced in Bardeen (1988) and extended into a multicomponent medium and the analysis in a class of generalized gravity theory (Hwang 1991a, hereafter H1). In H1 the so-called frame-choosing freedom was completely generalized, thus allowing us to write the full set of the equations independent of the frame choice. One particularly simplifying aspect of the method—clearly presented in Bardeen (1988)—is that, in dealing with cosmological perturbation analysis based on the homogeneous and isotropic background "space," the spatial gauge mode has no significance. One can write the full set of the equations without involving the spatially gauge-dependent variables. Thus the equation can be written in terms of variables which depend only on temporal gauge transformation (time slice or spacelike hypersurface condition). Letting each perturbed variable in the equation equal to zero corresponds to choosing a particular gauge.

This gauge nonspecific method has been applied by deriving complete solutions in an ideal-fluid case (Hwang 1993a, hereafter H2). In H2 six different gauges were identified, which include most of the conventionally used gauge choices including the synchronous gauge. As another application, in this paper, the method is applied to the medium dominated by a minimally coupled scalar field. We again identify six different gauge choices. In this medium, the large- and small-scales (depending on whether the spatial gradient term appearing is negligible or dominating, respectively; see e.g., eqs. [26] and [37], etc.) asymptotic solutions were known in the zero-shear gauge (Mukhanov 1985). Using the gauge nonspecific method the general asymptotic solutions in all the other gauges are derived for general scalar field potential. Large-scale conserved quantities, independent of the changing equation of state [i.e., for general pressure $p(\mu)$], are identified. Also shown are the closed form of the perturbed scalar field equation in each gauge choice. We also show that deriving the asymptotic solution in each different gauge choice is easily available. The equations and the solutions are compared to the corresponding ideal fluid case where the similarity between them is distinguished in the large scale. Similarly as in the ideal fluid case, exact solutions can be derived when the pressure to density ratio remains constant. This condition leads to a power-law type expansion stage which is supported by exponential type scalar field potential.

In § 2, we present the fundamental equations we need for perturbation analysis of the scalar field in Friedmann-Lemaître-

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Robertson-Walker (FLRW) spacetime and identify six different gauge choices. We also present our semiclassical method of treating the perturbed part as the quantum operators. In § 3, we present the perturbed equations of motion for the scalar field and the general asymptotic solutions available in six fundamental gauge choices. Solutions are presented in tabular form and the large-scale conserved quantities are identified. In § 4 we present the exact solutions available in the exponential potential background scalar field. We also present the solutions, both exact and asymptotic, in tabular form. Exact de Sitter background leads to a degenerate case which is separately treated in § 5. From our thorough analysis we will find out that the case of uniform-curvature gauge is distinguishing. The equation is most simple, and, furthermore, an interesting cancellation occurs in power-law expansion case such that the perturbed equation of motion exactly reduces to the case without considering the metric perturbations. More comments regarding this gauge is made in § 6. Section 7 is a discussion.

As a unit we set $c \equiv 1$. We keep the gravitational constant as $\gamma \equiv 8\pi G$.

2. FORMULATIONS

Since the basic formalism of the perturbation analysis, including the scalar field, was fully developed in H1, we mainly quote equations derived in that paper.

2.1. *Equations for Minimally Coupled Scalar Field*

Lagrangian in this case can be written as

$$L = \frac{1}{2\gamma} R - \left[\frac{1}{2}\phi_{,a}\phi^{;a} + V(\phi)\right], \tag{1}$$

where R and ϕ are the Ricci curvature and the scalar field, respectively; the semicolon indicates the covariant derivative based on the spacetime metric. The covariant form of the fluid quantities, based on a four velocity u^a are (we write them to linear order; see eqs. [63]–[65] of H1; for fully covariant form, see eq. [22] of Hwang 1990)

$$\mu = \tfrac{1}{2}\dot{\phi}^2 + V\,, \qquad p = \tfrac{1}{2}\dot{\phi}^2 - V\,, \qquad q_a = -\dot{\phi} h_a^b \phi_{,b}\,, \qquad \pi_{ab} = 0\,, \tag{2}$$

where q_a and π_{ab} are the energy flux vector and the anisotropic pressure tensor; an overdot indicates a derivative with respect to the proper time, following the u^a vector, e.g., $\dot{\phi} \equiv \phi_{,a} u^a$; and $h_{ab} \equiv g_{ab} + u_a u_b$ is a projection tensor projecting a quantity orthogonal to u^a. (This is a part of the analysis employing the covariant equations; one can find a more complete flavor of the advantage of using the covariant equations in cosmological perturbation analysis in Hwang & Vishniac 1990; Hwang 1991b and H1.)

Background equations for FLRW spacetime are (H1, eq. [21]):

$$\dot{\mu} = -3H(\mu + p)\,, \qquad H^2 = \frac{1}{3}(\gamma\mu + \Lambda) - \frac{K}{a^2}\,, \tag{3}$$

where the second equation is called as the Friedmann equation; $a(t)$ indicates the cosmic scale factor. $H(t) \equiv \dot{a}/a$; $\mu(t)$, $p(t)$, Λ and K indicate the energy density, the pressure, the cosmological constant, and the sign of three-space curvature, respectively; an overdot is the coordinate time derivative of the FLRW metric. From equation (3) follows

$$\dot{H} = -\frac{\gamma}{2}(\mu + p) + \frac{K}{a^2}\,. \tag{4}$$

Background fluid quantities are

$$\mu = \tfrac{1}{2}\dot{\phi}^2 + V\,, \qquad p = \tfrac{1}{2}\dot{\phi}^2 - V\,, \qquad q_a = 0 = \pi_{ab}\,. \tag{5}$$

To distinguish the background quantities from the covariant quantities in equation (2) we may need to denote the background quantities with bars, e.g., $\mu = \bar{\mu} + \epsilon$; we omit the bar unless it is necessary. Background equations (eqs. [3]–[4]) become

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0\,, \qquad H^2 = \frac{1}{3}\left[\gamma\left(\frac{1}{2}\dot{\phi}^2 + V\right) + \Lambda\right] - \frac{K}{a^2}\,, \qquad \dot{H} = -\frac{\gamma}{2}\dot{\phi}^2 + \frac{K}{a^2}\,, \tag{6}$$

where the first equation corresponds to the equation of motion.

The suggested form of perturbed equations in gauge nonspecific formulation are equations (22)–(28) of H1. By accepting that equations, gauge transformation properties (§ 2.2 of H1) and equation of state (eq. [7] for the scalar field) this paper can be considered as self-contained. Perturbed fluid quantities are (see eq. [67] of H1):

$$\epsilon = \dot{\phi}\,\delta\dot{\phi} - \dot{\phi}^2\alpha + V_{,\phi}\,\delta\phi\,, \qquad \pi = \dot{\phi}\,\delta\dot{\phi} - \dot{\phi}^2\alpha - V_{,\phi}\,\delta\phi\,, \qquad \Psi = -\dot{\phi}\,\delta\phi\,, \qquad \sigma = 0\,. \tag{7}$$

(Considering that $\mu = \bar{\mu} + \epsilon$ and $p = \bar{p} + \pi$, ϵ and π in eq. [7] can be derived by perturbing eq. [2], and considering the fact that the overdot in eq. [2] is based on a proper time derivative which differs from the coordinate time derivative used in eqs. [5]–[7] by the perturbed part of lapse function α.) The perturbed variables are α (perturbed lapse function), φ (perturbed three-space curvature), χ (shear), and κ (perturbed expansion scalar), which are the perturbed metric variables, whereas ϵ (perturbed energy density), π (perturbed pressure), Ψ (perturbed flux, or velocity), and σ (anisotropic pressure) are the perturbed matter variables.

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Combining equations (22)–(27) of H1 with equation (7) we can derive

$$\kappa = -3\dot{\varphi} + 3H\alpha + \frac{k^2}{a^2}\chi , \tag{8}$$

$$-H\kappa + \frac{k^2 - 3K}{a^2}\varphi = \frac{\gamma}{2}(\dot{\phi}\,\delta\dot{\phi} - \dot{\phi}^2\alpha + V_{,\phi}\,\delta\phi) , \tag{9}$$

$$\kappa - \frac{k^2 - 3K}{a^2}\chi = \frac{3\gamma}{2}\dot{\phi}\,\delta\phi , \tag{10}$$

$$\alpha + \varphi = \dot{\chi} + H\chi , \tag{11}$$

$$\dot{\kappa} + 2H\kappa = \left(\frac{k^2 - 3K}{a^2} - \frac{\gamma}{2}\dot{\phi}^2\right)\alpha + \gamma(2\dot{\phi}\,\delta\dot{\phi} - V_{,\phi}\,\delta\phi) , \tag{12}$$

$$\delta\ddot{\phi} + 3H\,\delta\dot{\phi} + \left(\frac{k^2}{a^2} + V_{,\phi\phi}\right)\delta\phi = \dot{\phi}(\kappa + \dot{\alpha}) - (3H\dot{\phi} + 2V_{,\phi})\alpha , \tag{13}$$

which are the basic equations we need in treating the perturbed FLRW space, which is filled with a minimally coupled scalar field. The last equation corresponds to the equation of motion. Equation (28) of H1 is identically satisfied.

We can estimate the dimensions of the background and the perturbed variables in our unit system where $c \equiv 1$. Since $[\gamma] = [8\pi G] = [M^{-1}T^2]$, where M and T indicate dimensions of the mass (energy) and the time (length), respectively, we have

$$[\mu] = [p] = [\epsilon] = [\pi] = [MT^{-4}] , \qquad [\Psi] = [MT^{-3}] ,$$
$$[T\kappa] = [\varphi] = [\alpha] = [T^{-1}\chi] = 0 , \qquad [\phi] = [\delta\phi] = [M^{1/2}T^{-1}] , \qquad [V] = [MT^{-4}] . \tag{14}$$

From these, it follows that κ/H, φ, α, and $\gamma\Psi/H$ are dimensionless.

In the following we *assume* $K = 0 = \Lambda$. Thus, from equation (6), we have

$$H^2 = \frac{\gamma}{3}\mu , \qquad \dot{H} = -\frac{\gamma}{2}(\mu + p) = -\frac{\gamma}{2}\dot{\phi}^2 . \tag{15}$$

We will use the following conventions:

$$w \equiv \frac{p}{\mu} , \qquad c_s^2 \equiv \frac{\dot{p}}{\dot{\mu}} = 1 + \frac{2}{3H}\frac{V_{,\phi}}{\dot{\phi}} . \tag{16}$$

2.2. *Gauge Choices*

From equations (7)–(13) we can summarize the fundamental gauge choices as the following: zero-shear gauge ($\chi \equiv 0$; ZSG), comoving gauge ($\Psi \equiv 0$; CG), synchronous gauge ($\alpha \equiv 0$; SG), uniform-expansion gauge ($\kappa \equiv 0$; UEG), uniform-curvature gauge ($\varphi \equiv 0$; UCG), and uniform-density gauge ($\epsilon \equiv 0$; UDG). The ZSG is also often called as a longitudinal gauge. The CG condition implies $\delta\phi = 0$ (eq. [7]) thus may also be called a uniform-field gauge. We can also introduce a uniform-pressure gauge ($\pi \equiv 0$). The spacelike hypersurface of this gauge may be more complicated than the UDC hypersurface. In this paper we do not consider this gauge.

We note that except for the SG, each of the other gauge conditions *completely removes* the gauge mode from the analysis which may be an apparent advantage over the SG choice (see § 2.2 and 3.1 of H1). However, the SG choice, despite its shortcoming of not completely removing the gauge mode, still allows a simple form of fluid quantities (see eq. [7]) which becomes more apparent in the multicomponent fluids or field cases (see eqs. [35] and [66] of H1).

2.3. *Quantization*

We restore all the perturbed variables ($\delta\phi$, φ, κ, α, χ, ϵ, π, Ψ, etc., in eqs. [7]–[13]) into physical space and replace each variable by a quantum (Heisenberg representation) operator. Thus we are splitting the scalar field and the matter and the metric variables as

$$\phi(\boldsymbol{x}, t) = \bar{\phi}(t) + \delta\hat{\phi}(\boldsymbol{x}, t) , \qquad \mu(\boldsymbol{x}, t) = \bar{\mu}(t) + \hat{\epsilon}(\boldsymbol{x}, t) , \quad \text{etc.} \tag{17}$$

We neglect the still subtle issue about precise meaning of these splitting in quantum mechanical context (see Guth & Pi 1985). For $\delta\hat{\phi}(\boldsymbol{x}, t)$ based on pure background metric, not considering any metric perturbations (eq. [26]), we require the equal time commutation relation to hold

$$[\delta\hat{\phi}(\boldsymbol{x}, t), \delta\dot{\hat{\phi}}(\boldsymbol{x}', t)] = ia^{-3}\delta^3(\boldsymbol{x} - \boldsymbol{x}') , \tag{18}$$

where $\delta^3(\boldsymbol{x})$ indicates a δ-function. Later we will see that (1) in the small-scale asymptote, (2) in the de Sitter background, (3) in $G \to 0$ limit, cases 1–3 apply to most of the gauge choices, and also (4) in power-law expansion background in the UCG, the equation for $\delta\phi$ can be identified to the one based on pure background metric (eq. [26]). Thus, at least in such cases, the relation in equation (18)

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may still hold even considering the metric perturbations. Since we are considering a *flat* background space, we may expand $\delta\hat{\phi}(\boldsymbol{x}, t)$ in the following mode expansion

$$\delta\hat{\phi}(\boldsymbol{x}, t) = \int \frac{d^3k}{(2\pi)^{3/2}} [\hat{a}_{\boldsymbol{k}}\, \delta\phi_{\boldsymbol{k}}(t) e^{i\boldsymbol{k}\cdot\boldsymbol{x}} + \hat{a}^{\dagger}_{\boldsymbol{k}}\, \delta\phi^{*}_{\boldsymbol{k}}(t) e^{-i\boldsymbol{k}\cdot\boldsymbol{x}}] , \tag{19}$$

where $\hat{a}_{\boldsymbol{k}}$ and $\hat{a}^{\dagger}_{\boldsymbol{k}}$ satisfy standard commutation relation:

$$[\hat{a}_{\boldsymbol{k}}, \hat{a}_{\boldsymbol{k}'}] = 0 , \qquad [\hat{a}^{\dagger}_{\boldsymbol{k}}, \hat{a}^{\dagger}_{\boldsymbol{k}'}] = 0 , \qquad [\hat{a}_{\boldsymbol{k}}, \hat{a}^{\dagger}_{\boldsymbol{k}'}] = \delta^3(\boldsymbol{k} - \boldsymbol{k}') , \tag{20}$$

and $\delta\phi_{\boldsymbol{k}}(t)$ is the mode function, a complex solution of the classical mode evolution equation. In order that equation (20) be in accord with equation (18) the mode function $\delta\phi_{\boldsymbol{k}}(t)$ should follow the Wronskian condition

$$\delta\phi_{\boldsymbol{k}}\, \delta\dot{\phi}^{*}_{\boldsymbol{k}} - \delta\phi^{*}_{\boldsymbol{k}}\, \delta\dot{\phi}_{\boldsymbol{k}} = i a^{-3} . \tag{21}$$

The normalization conditions in equations (18) and (21) are imposed on $\delta\hat{\phi}(\boldsymbol{x}, t)$ and $\delta\phi_{\boldsymbol{k}}(t)$, which are *based* on the background metric without considering the metric perturbations; since our complete (considering the full metric perturbations) equations reduce to equation (26) in the cases 1–4, this restriction about quantization condition is sufficient in this paper. Proper analysis may need Lagrangian and Hamiltonian treatments where the same set of equations will be derived (see Mukhanov, Feldman, & Brandenberger 1992).

In classical context, our set of the complete perturbation equations (eqs. [7]–[13]) are written for the Fourier component of each variable; we are considering the flat background metric. Thus, for example,

$$\delta\phi = \delta\phi_{\boldsymbol{k},\mathrm{F}}(t) , \qquad \varphi = \varphi_{\boldsymbol{k},\mathrm{F}}(t) , \quad \text{etc.} \tag{22}$$

where

$$\delta\phi_{\boldsymbol{k},\mathrm{F}}(t) = \frac{1}{\sqrt{\mathrm{Vol}}} \int d^3x e^{i\boldsymbol{k}\cdot\boldsymbol{x}} \delta\phi(\boldsymbol{x}, t) , \quad \text{etc.} \tag{23}$$

According to our canonical quantization explained above, we restore the perturbed variables into the physical space, regard each variable as a quantum (Heisenberg) operator, and mode expand as in equation (19). Thus for $\delta\hat{\phi}$ we expand as in equation (19), and for $\hat{\varphi}$ we expand as

$$\hat{\varphi}(\boldsymbol{x}, t) = \int \frac{d^3k}{(2\pi)^{3/2}} [\hat{a}_{\boldsymbol{k}}\, \varphi_{\boldsymbol{k}}(t) e^{i\boldsymbol{k}\cdot\boldsymbol{x}} + \hat{a}^{\dagger}_{\boldsymbol{k}}\, \varphi^{*}_{\boldsymbol{k}}(t) e^{-i\boldsymbol{k}\cdot\boldsymbol{x}}] , \tag{24}$$

and similarly for $\hat{\kappa}$, $\hat{\alpha}$, $\hat{\chi}$, $\hat{\epsilon}$, $\hat{\pi}$, $\hat{\Psi}$, etc. (e.g., Salopek, Bond, & Bardeen 1989). Then the same equations (eqs. [7]–[13] will hold, with each variable regarded as a mode function counterpart. We will employ this identical form of the equations in simplifying the notation; we could interpret all the variables in equations (7)–(13) to be the mode function variables instead of the Fourier component; we will often omit subindex $\boldsymbol{k}$ in equations. Thus we can consider

$$\delta\phi \equiv \delta\phi(\boldsymbol{k}, t) = \delta\phi_{\boldsymbol{k}}(t) , \qquad \varphi \equiv \varphi(\boldsymbol{k}, t) = \varphi_{\boldsymbol{k}}(t) , \quad \text{etc.} \tag{25}$$

However, imposition of the commutation (quantization) relations for the quantum scalar field and metric variables is yet *reserved* at this stage (see eq. [45] and § 3.7). (In our previous treatment, as often in the literature, the perturbation variables in eqs. [7]–[13] were regarded as the physical space ones, which is indeed allowed by simply changing k^2 into $-\nabla^{(3)2}$.)

2.4. *Scalar Field Perturbations Based on Background Metric*

By neglecting the metric fluctuations, we have (eq. [13])

$$\delta\ddot{\phi} + 3H\delta\dot{\phi} + (a^{-2}k^2 + V_{,\phi\phi})\delta\phi = 0 . \tag{26}$$

This equation is consistent with the rest of the equations (eqs. [9], [10], and [12]) if we consider the limit of negligible gravitational coupling constant, i.e., $\gamma \to 0$.

Later we will show that in some inflationary expansion stages the fully (including metric perturbations) perturbed equation in some particular gauge become *identical* to equation (26). This is the case for the UCG in the power-law expanding stage (§§ 3.5 and 6) and in general for an exponential expansion stage (§ 5). Also in the small-scale, the fully perturbed equations for $\delta\phi$ in most of the gauge choice coincide with the same limit of equation (26); these are the cases for the ZSG, the UEG, the SG, and the UCG (see eq. [83]). It is amusing to notice that the fully perturbed cases have more internal consistencies (see § 6). Actually, that must be the case since perturbed scalar field will necessarily simultaneously excite the metric field.

In the following we summarize the mode function solutions in two special expansion stages.

1. *Exponential expansion.*—The exponential expansion stage is characterized by

$$a = a_0 e^{H(t - t_0)} , \qquad H = \text{constant} . \tag{27}$$

Assuming free massive scalar field, $V = \frac{1}{2}m^2\phi^2$, equation (26) can be solved (procedure described in § 2.3 is understood) to give (Bunch & Davies 1978)

$$\delta\phi_{\boldsymbol{k}}(\eta) = \frac{\sqrt{\pi}}{2} H\eta^{3/2}[c_1(k)H^{(1)}_{\nu}(k\eta) + c_2(k)H^{(2)}_{\nu}(k\eta)] , \qquad \nu \equiv \sqrt{\frac{9}{4} - \frac{m^2}{H^2}} , \tag{28}$$

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where η is a conformal time ($d\eta = a^{-1}\,dt$) which in exponential expansion background becomes $\eta = -(aH)^{-1}$. The coefficients $c_1(k)$ and $c_2(k)$ are arbitrary functions of k which are normalized according to equation (21) as

$$|c_2(k)|^2 - |c_1(k)|^2 = 1 . \tag{29}$$

Imposing the quantization condition of equation (21) does not completely fix the coefficient which is related to the choice of the vacuum state. The adiabatic vacuum choice corresponds to choosing $c_2(k) \equiv 1$ and $c_1(k) \equiv 0$. For massless field (thus $\nu = 3/2$) we have

$$\delta\phi_{\mathbf{k}}(\eta) = c_1(k)\frac{1}{\sqrt{2k}a}\left(1 + \frac{i}{k\eta}\right)e^{ik\eta} + c_2(k)\frac{1}{\sqrt{2k}a}\left(1 - \frac{i}{k\eta}\right)e^{ik\eta} . \tag{30}$$

2. *Power-law expansion.*—Power-law expansion stage is characterized by

$$a = (\eta/\eta_0)^{2/(1+3w)}a_0 , \tag{31}$$

where $w \equiv p/\mu$ is assumed to be a constant with $w < -\frac{1}{3}$. *Assuming* $V(\phi) = 0$, equation (26) can be solved (Ford & Parker 1977; § 7.2 of Birrell & Davies 1982; Pathinayake & Ford 1988; Sahni 1988) as

$$\delta\phi_{\mathbf{k}}(\eta) = -\frac{\sqrt{\pi\eta_0}}{2a_0}\left(\frac{\eta}{\eta_0}\right)^{\nu}[c_1(k)H_\nu^{(1)}(k\eta) + c_2(k)H_\nu^{(2)}(k\eta)] , \qquad \nu \equiv \frac{3(w-1)}{2(3w+1)} , \tag{32}$$

where

$$\frac{\sqrt{\eta}}{a} = \frac{\sqrt{\eta_0}}{a_0}\left(\frac{\eta}{\eta_0}\right)^{\nu} . \tag{33}$$

The coefficients $c_1(k)$ and $c_2(k)$ are normalized according to equation (29). In the exponential expansion limit ($w \to -1$, $\nu \to 3/2$), equation (32) approaches the exponential case of equation (28).

From the mode function solution we can derive power spectrum (based on vacuum expectation value), which is defined as (Hwang 1993b)

$$\Delta\hat{\phi}(k, t) \equiv \sqrt{\frac{k^3}{2\pi^2}}\,|\delta\phi_{\mathbf{k}}(t)| . \tag{34}$$

In the large-scale, and taking the adiabatic vacuum, we have, for exponential expansion (from eq. [30])

$$\Delta\hat{\phi}(k, t) = \frac{H}{2\pi} , \tag{35}$$

and for the power-law expansion (from eq. [32]):

$$\Delta\hat{\phi}(k, t) = \frac{\Gamma(\nu)}{\pi^{3/2}a|\eta|}\left(\frac{k|\eta|}{2}\right)^{\overline{3/2-\nu}} . \tag{36}$$

3. PERTURBED EQUATIONS AND ASYMPTOTIC SOLUTIONS

3.1. *Zero-Shear Gauge*

The ZSG imposes $\chi \equiv 0$. From equations (8)–(13) we can derive

$$\delta\ddot{\phi} + \left(3H + \frac{\ddot{\phi} + H\dot{\phi}}{\gamma^{-1}k^2/a^2 - \dot{\phi}^2/2}\,\dot{\phi}\right)\delta\dot{\phi} + \left(\frac{k^2}{a^2} + V_{,\phi\phi} + 4\dot{H} - \frac{\dddot{\phi} + H\ddot{\phi}}{\gamma^{-1}k^2/a^2 - \dot{\phi}^2/2}\,\dot{\phi}\right)\delta\phi = 0 . \tag{37}$$

This equation is modified in both $\delta\dot{\phi}$ and $\delta\phi$ terms compared to the equation based on the background metric (eq. [26]). In the limit of $G \to 0$ we recover equation (26). In the small-scale equation (37) approaches equation (26) in the same scale.

From equations (8)–(13) we can also derive

$$\ddot{\varphi} + \left(H - 2\frac{\ddot{\phi}}{\dot{\phi}}\right)\dot{\varphi} + \left(\frac{k^2}{a^2} + 2\dot{H} - 2H\frac{\ddot{\phi}}{\dot{\phi}}\right)\varphi = \frac{\dot{H}}{H}\left[\frac{H^2}{a\dot{H}}\left(\frac{a}{H}\varphi\right)^{\cdot}\right]^{\cdot} + \frac{k^2}{a^2}\varphi = 0 . \tag{38}$$

Using the fluid variables in equation (16), equation (38) can also be written as

$$\ddot{\varphi} + (4 + 3c_s^2)H\dot{\varphi} + \left[\frac{k^2}{a^2} + \gamma\mu(c_s^2 - w)\right]\varphi = 0 . \tag{39}$$

This equation can be compared with the fluid case. From equations (22)–(28) of H1 we have

$$\ddot{\varphi} + (4 + 3c_s^2)H\dot{\varphi} + \left[c_s^2\frac{k^2}{a^2} + \gamma\mu(c_s^2 - w)\right]\varphi = -\frac{\gamma}{2}e . \tag{40}$$

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where $\pi \equiv c_s^2 + e$, and we neglect the anisotropic pressure ($\sigma = 0$). Indeed, one can show that for a minimally coupled scalar field:

$$e = \frac{2}{\gamma}(1 - c_s^2)\frac{k^2}{a^2}\varphi , \qquad \sigma = 0 , \tag{41}$$

thus for a scalar field, equations (39) and (40) are identical. Compared to the ideal fluid (meaning $e = 0 = \sigma$) equation, minimally coupled scalar field equation can be considered as replacing c_s^2 in front of $(k/a)^2$ term to unity. Similar difference occur in the fluid formulation of the multicomponent scalar field case. In that case the caused exchange of terms (like $c_s^2 k^2 \to k^2$) occur in a more striking way (see § 3.3.2 of H1; especially compare eqs. [38] and [97] considering eq. [96] in that paper). Thus, in the large scale, neglecting the $(k/a)^2$ term, the solutions for an ideal fluid and a scalar field will show identical behavior (in this gauge).

Thus in the large scale, also regarding the entropic perturbations e is negligible in this scale in the fluid case, from the second steps of equation (38), we can derive general integral form solution as

$$\varphi = -C\frac{H}{a}\int^2 a\frac{\dot{H}}{H^2}\,dt = C\frac{H}{a}\int^t a\left(\frac{1}{H}\right)^{\cdot} dt = C\left(1 - \frac{H}{a}\int^t a\,dt\right), \tag{42}$$

where the upper and the lower bound of integration indicate the growing and the decaying modes, respectively; $C(\boldsymbol{k})$ is a constant coefficient for the growing mode. Equation (42) is valid for general $p(\mu)$, thus for general $V(\phi)$.

In the *large scale*, thus we neglect $(k/a)^2$ (second-order spatial gradient) terms, equation (37) can be written in a compact form as

$$\dot{\phi}\left[\frac{1}{a}\left(\frac{a}{\dot{\phi}}\delta\phi\right)^{\cdot}\right]^{\cdot} = 0 . \tag{43}$$

Thus we have

$$\frac{\delta\phi}{\dot{\phi}} = -C\frac{1}{a}\int^t a\,dt , \tag{44}$$

where the lower bound of integration shows the behavior of the decaying mode and the coefficient is matched with the solution in equation (42). From this single solution of equation (44) follows all the rest of the solutions in the same gauge, and from them we can derive solutions in every other gauge; the method will be described in each specific gauge case, separately. In Table 1, $\varphi = C$ indicates that the dominating decaying mode has canceled. It is very important to note that C is a constant that is independent of the changing background equation of state, i.e., for general $p(\mu)$ or $V(\phi)$. That is, as long as the scale is in the large scale we have integral form solutions where C is a constant independent of the relation between μ and p, thus between V and ϕ. Information about the spatial structure is contained in $C(\boldsymbol{x})$ and the coefficient of the decaying mode, and is preserved.

In the *small scale*, equation (38) can be solved, to WKB order, to give

$$\varphi = \frac{i\gamma\dot{\phi}}{(2k)^{3/2}}\left[c_1(k)e^{ik\eta} - c_2(k)e^{-ik\eta}\right] . \tag{45}$$

The time-independent coefficients are chosen so that $\delta\phi$ coincide with the normalization in equation (30) (see eq. [83]) with $c_1(k)$ and $c_2(k)$ following equation (29); as noted, in the small-scale equation (37) approaches equation (26) without the potential term, thus justifying using the normalization condition (following from the quantization condition) for $\delta\phi$ based on pure background metric (see § 3.7). More justification of using the solution derived in § 2.4 will be presented in § 6 in the power law and in § 5 in the exponential expansion cases. From this solution in equation (45) follow the solutions for all the rest of the variables. The solutions for small scale are presented in Table 2.

We note that the large- and the small-scale solutions in equations (44) and (45), and their corresponding full solutions presented in Tables 1 and 2, are generally applicable to wide range of background space dynamics. That is, if we choose specific background space dynamics it will imply specifying $a(t)$, thus $H(t)$, $\phi(t)$, $V[\phi(t)]$, etc. In this sense the asymptotic solutions presented in Tables 1 and 2 are general.

Introducing [Mukhanov 1985, 1988 (Notice different conventions, their $\Phi = \Psi$ corresponds to our $-\varphi_\chi = \alpha_\chi$; $\varphi_\chi \equiv \varphi - H\chi$ which becomes φ in the ZSG where $\chi \equiv 0$, and equivalently it becomes $-H\chi$ in the UCG; see §§ 3.7–3.8)],

$$u \equiv -\frac{\varphi_\chi}{\dot{\phi}} , \tag{46}$$

equation (38) can be written as

$$u'' + \left[k^2 - \frac{(1/z)''}{1/z}\right]u = z[z^{-2}(zu)']' + k^2 u = 0 , \qquad z \equiv \frac{\phi'}{H} , \tag{47}$$

where a prime denotes a derivative based on conformal time ($d\eta = a^{-1}\,dt$).

3.2. *Comoving Gauge*

The CG imposes $\Psi \equiv 0$, which implies $\delta\phi = 0$ (eq. [7]). Thus we may also refer to this gauge as the uniform-field gauge. Naturally, we do not have an equation for $\delta\ddot{\phi}$. Instead, in this CG, an equation for $\delta \equiv \epsilon/\mu$ is often convenient. From equations (7),

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TABLE 1

LARGE-SCALE ASYMPTOTIC SOLUTIONS IN THE INTEGRAL FORMS

Variables/Gauges	$\Psi = 0 = \delta\phi$ (Comoving)	$\chi = 0$ (Zero Shear)	$\kappa = 0$ (Uniform Expansion)	$\alpha = 0$ (Synchronous)
$\gamma\Psi/H$	0	$-2C\frac{\dot H}{\dot a}\int^t a\,dt$	$\frac{2}{3}C\frac{H}{a}\int^t a\,dt\left(\frac{k}{aH}\right)^2$	$\frac{2}{H}\dot\varphi = \mathcal{O}\left[\left(\frac{k}{aH}\right)^2 C\right] + 2g\frac{\dot H}{H}$
$H\,\delta\phi/\dot\phi$	0	$-C\frac{H}{a}\int^t a\,dt$	$\frac{1}{3}C\frac{H^3}{a\dot H}\int^t a\,dt\left(\frac{k}{aH}\right)^2$	$\frac{H}{\dot H}\dot\varphi = \mathcal{O}\left[\left(\frac{k}{aH}\right)^2 C\right] + gH$
$H\chi$	$C\frac{H}{a}\int^t a\,dt$	0	$C\frac{H}{a}\int^t a\,dt$	$C\frac{H}{a}\int^t a\,dt + gH$
κ/H	$C\frac{H}{a}\int^t a\,dt\left(\frac{k}{aH}\right)^2$	$3C\frac{\dot H}{\dot a}\int^t a\,dt$	0	$-\frac{3}{H}\dot\varphi = \mathcal{O}\left[\left(\frac{k}{aH}\right)^2 C\right] - 3g\frac{\dot H}{H}$
α	$C\frac{H^2}{\dot H}\left(1 - \frac{H}{a}\int^t a\,dt\right)\left(\frac{k}{aH}\right)^2$	$-C\left(1 - \frac{H}{a}\int^t a\,dt\right)$	$\frac{2}{3}C\frac{H^2}{\dot H}\left(2 + \frac{V_{,\phi}}{\dot\phi}\frac{1}{a}\int^t a\,dt\right)\left(\frac{k}{aH}\right)^2$	0
φ	C	$C\left(1 - \frac{H}{a}\int^t a\,dt\right)$	C	$C + gH$
ϵ/μ	$\frac{2}{3}C\left(1 - \frac{H}{a}\int^t a\,dt\right)\left(\frac{k}{aH}\right)^2$	$-2C\frac{\dot H}{\dot a}\int^t a\,dt$	$\frac{2}{3}C\left(\frac{k}{aH}\right)^2$	$\frac{2}{H}\dot\varphi = \mathcal{O}\left[\left(\frac{k}{aH}\right)^2 C\right] + 2g\frac{\dot H}{H}$
π/μ	$\frac{2}{3}C\left(1 - \frac{H}{a}\int^t a\,dt\right)\left(\frac{k}{aH}\right)^2$	$-2Cc_s^2\frac{\dot H}{\dot a}\int^t a\,dt$	$\frac{2}{3}C\left(1 + \frac{2}{3}\frac{V_{,\phi}}{\dot\phi}\frac{1}{a}\int^t a\,dt\right)\left(\frac{k}{aH}\right)^2$	$c_s^2\frac{2}{H}\dot\varphi = \mathcal{O}\left[\left(\frac{k}{aH}\right)^2 C\right] + 2gc_s^2\frac{\dot H}{H}$

Variables/Gauges	$\varphi = 0$ (Uniform Curvature)	$\epsilon = 0$ (Uniform Density)
$\gamma\Psi/H$	$-2C\frac{\dot H}{H^2}$	$-\frac{2}{3}C\left(1 - \frac{H}{a}\int^t a\,dt\right)\left(\frac{k}{aH}\right)^2$
$H\,\delta\phi/\dot\phi$	$-C$	$-\frac{1}{3}C\frac{H^2}{\dot H}\left(1 - \frac{H}{a}\int^t a\,dt\right)\left(\frac{k}{aH}\right)^2$
$H\chi$	$-C\left(1 - \frac{H}{a}\int^t a\,dt\right)$	$C\frac{H}{a}\int^t a\,dt$
κ/H	$3C\frac{\dot H}{H^2}$	$C\left(\frac{k}{aH}\right)^2$
α	$C\frac{\dot H}{H^2}$	$\frac{1}{3}C\left[1 - 2\frac{V_{,\phi}}{\dot\phi}\frac{H}{\dot H}\left(1 - \frac{H}{a}\int^t a\,dt\right)\right]\left(\frac{k}{aH}\right)^2$
φ	0	C
ϵ/μ	$-2C\frac{\dot H}{H^2}$	0
π/μ	$-2Cc_s^2\frac{\dot H}{H^2}$	$-\frac{4}{9}C\frac{V_{,\phi}}{H\dot\phi}\left(1 - \frac{H}{a}\int^t a\,dt\right)\left(\frac{k}{aH}\right)^2$

NOTES.—Solutions are valid for general $V(\phi)$. Except for the solutions in the SG, the upper and the lower bounds of integration indicate the growing and the decaying modes. No appearance of integration in the solution implies that the dominating decaying mode has canceled. See § 3.4 for the SG case; g is a coefficient of the gauge mode.

(12), and (13) we can derive

$$\ddot\delta + \left(5H + 4\frac{\dot H}{H} - 2\frac{\ddot\phi}{\dot\phi}\right)\dot\delta + \left(\frac{k^2}{a^2} + 6H^2 + 14\dot H + 2\frac{\dot H^2}{H^2} - 6H\frac{\ddot\phi}{\dot\phi}\right)\delta = \frac{\dot H}{H^3 a^2}\left[\frac{H^2}{a\dot H}(a^3 H\delta)^{\cdot}\right]^{\cdot} + \frac{k^2}{a^2}\delta = 0\,. \quad (48)$$

In terms of the fluid variables (eq. [16]) this equation can be written as

$$\ddot\delta + (2 + 3c_s^2 - 6w)H\dot\delta + \left[\frac{k^2}{a^2} + \gamma\mu\left(\frac{3}{2}w^2 + 3c_s^2 - 4w - \frac{1}{2}\right)\right]\delta = 0\,. \quad (49)$$

The complete equation for the fluid case follows from equations (22)–(28) of H1 as

$$\ddot\delta + (2 + 3c_s^2 - 6w)H\dot\delta + \left[c_s^2\frac{k^2}{a^2} + \gamma\mu\left(\frac{3}{2}w^2 + 3c_s^2 - 4w - \frac{1}{2}\right)\right]\delta = -\frac{k^2}{a^2}\frac{e}{\mu}\,, \quad (50)$$

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TABLE 2

SMALL-SCALE (WKB) ASYMPTOTIC SOLUTION

Variables/Gauges	$\Psi = 0 = \delta\phi$ (Comoving)	$\chi = 0$ (Zero Shear)	$\kappa = 0$ (Uniform Expansion)	$\alpha = 0$ (Synchronous)
Ψ	0	$-\frac{\dot{\phi}}{\sqrt{2k}a}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$-\frac{\dot{\phi}}{\sqrt{2k}a}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$-\frac{\dot{\phi}}{\sqrt{2k}a}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$
$\delta\phi$	0	$\frac{1}{\sqrt{2k}a}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{1}{\sqrt{2k}a}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{1}{\sqrt{2k}a}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$
χ	$-\frac{1}{\sqrt{2k}a\dot{\phi}}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	0	$-\frac{3\gamma a\dot{\phi}}{2^{3/2}k^{5/2}}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{\gamma a\dot{\phi}}{2^{3/2}k^{5/2}}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$
κ	$-\frac{k^{3/2}}{\sqrt{2}a^3\dot{\phi}}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{3\gamma\dot{\phi}}{2^{3/2}\sqrt{k}a}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	0	$\frac{\sqrt{2}\gamma\dot{\phi}}{\sqrt{k}a}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$
α	$-\frac{i\sqrt{k}}{\sqrt{2}a^2\dot{\phi}}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$-\frac{i\gamma\dot{\phi}}{(2k)^{3/2}}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{i\sqrt{2}\gamma\dot{\phi}}{k^{3/2}}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	0
φ	$-\frac{H}{\sqrt{2k}a\dot{\phi}}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{i\gamma\dot{\phi}}{(2k)^{3/2}}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{i\gamma\dot{\phi}}{(2k)^{3/2}}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{i\gamma\dot{\phi}}{(2k)^{3/2}}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
ϵ	$\frac{i\sqrt{k}\dot{\phi}}{\sqrt{2}a^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{i\sqrt{k}\dot{\phi}}{\sqrt{2}a^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{i\sqrt{k}\dot{\phi}}{\sqrt{2}a^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{i\sqrt{k}\dot{\phi}}{\sqrt{2}a^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
π	$\frac{i\sqrt{k}\dot{\phi}}{\sqrt{2}a^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{i\sqrt{k}\dot{\phi}}{\sqrt{2}a^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{i\sqrt{k}\dot{\phi}}{\sqrt{2}a^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{i\sqrt{k}\dot{\phi}}{\sqrt{2}a^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$

Variables/Gauges	$\varphi = 0$ (Uniform Curvature)	$\epsilon = 0$ (Uniform Density)
Ψ	$-\frac{\dot{\phi}}{\sqrt{2k}a}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$-\frac{i\sqrt{k}\dot{\phi}}{3\sqrt{2}a^2H}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
$\delta\phi$	$\frac{1}{\sqrt{2k}a}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{i\sqrt{k}}{3\sqrt{2}a^2H}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
χ	$-\frac{i\gamma\dot{\phi}}{(2k)^{3/2}H}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{i\sqrt{k}}{3\sqrt{2}a^2H\dot{\phi}}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
κ	$-\frac{i\gamma\sqrt{k}\dot{\phi}}{2^{3/2}a^2H}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{ik^{5/2}}{3\sqrt{2}a^4H\dot{\phi}}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
α	$\frac{\gamma\dot{\phi}}{2^{3/2}\sqrt{k}aH}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$-\frac{k^{3/2}}{3\sqrt{2}a^3H\dot{\phi}}(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$
φ	0	$\frac{i\sqrt{k}}{3\sqrt{2}a^2\dot{\phi}}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
ϵ	$\frac{i\sqrt{k}\dot{\phi}}{\sqrt{2}a^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	0
π	$\frac{i\sqrt{k}\dot{\phi}}{\sqrt{2}a^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$-\frac{i\sqrt{2k}V_{,\phi}}{3a^2H}(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$

NOTES.—We neglect the gauge modes in the SG. The coefficients c_1 and c_2 indicate $c_1(k)$ and $c_2(k)$, respectively. If we impose quantization condition, these coefficients follow eq. (29).

where we neglect the anisotropic pressure. In fact, in this gauge we have $\pi = \epsilon$ (see eq. [7]), thus leading to $e = (1 - c_s^2)\epsilon$. Thus, equations (49) and (50) are identical for the scalar field case. Comparing equation (49) to the ideal fluid equation in the CG, the only difference is that the c_s^2 term in front of $(k/a)^2$ term becomes unity in a minimally coupled scalar field case. Using the Poisson-like relation (we restore K for this equation),

$$\frac{\gamma}{2}\epsilon_\Psi = \frac{k^2 - 3K}{a^2}\varphi_\chi , \tag{51}$$

which follows from equations (9) and (10). Equation (50) can be converted into equation (40); $\epsilon_\Psi \equiv \epsilon - 3H\Psi$, see §§ 3.7–3.8. Equation (51) can be regarded as an analogy of the Poisson's equation known in Newtonian gravity. Equation (50) for δ does have direct correspondence to the equation for Newtonian density fluctuations; thus, equation (40) for φ does have direct correspondence to the equation for Newtonian potential fluctuations (Hwang & Hyun 1994).

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In the large scale, and neglecting e term in that scale in the fluid case, we have an integral form solution (use the second step of eq. [48]):

$$\delta = C\frac{2}{3}\left(\frac{k}{aH}\right)^2\left(1 - \frac{H}{a}\int^t a\,dt\right), \tag{52}$$

where the coefficient is matched with the solutions in other gauges.

To derive solutions from the known ZSG ones, since

$$\frac{k^2}{a^2}\varphi_\chi = \frac{\gamma}{2}\epsilon_\Psi = \frac{\gamma}{2}(\dot\phi\,\delta\dot\phi - \ddot\phi\,\delta\phi - \dot\phi^2\alpha), \tag{53}$$

we can use

$$\alpha = \frac{1}{\dot H}\frac{k^2}{a^2}\varphi_{\rm ZSG}. \tag{54}$$

From this relation we can derive α in this gauge. Starting from this solution we can derive all the rest of the variables in this gauge (see Tables 1 and 2). Of course, we can also start directly from the solution in equation (52).

3.3. *Uniform-Expansion Gauge*

The UEG imposes $\kappa \equiv 0$. From equations (12) and (13) we can derive

$$\left(1 - \frac{2\dot\phi^2}{\dot\phi^2/2 - \gamma^{-1}k^2/a^2}\right)\delta\ddot\phi + \left[2H + \frac{\dot\phi(12H\dot\phi + 7V_{,\phi})}{\dot\phi^2/2 - \gamma^{-1}k^2/a^2} + \frac{2\dot\phi^2(\dot\phi\ddot\phi + 2H\gamma^{-1}k^2/a^2)}{(\dot\phi^2/2 - \gamma^{-1}k^2/a^2)^2}\right]\delta\dot\phi$$
$$+\left[\frac{k^2}{a^2} + V_{,\phi\phi} + \frac{V_{,\phi\phi}\dot\phi^2 + V_{,\phi}(3H\dot\phi + 2\ddot\phi)}{\dot\phi^2/2 - \gamma^{-1}k^2/a^2} - \frac{\dot\phi V_{,\phi}(\dot\phi\ddot\phi + 2H\gamma^{-1}k^2/a^2)}{(\dot\phi^2/2 - \gamma^{-1}k^2/a^2)^2}\right]\delta\phi = 0. \tag{55}$$

This equation is modified in many ways compared to the equation based on the background metric (eq. [26]). In the limit of $G \to 0$ we still recover equation (26). In the small-scale, equation (55) approaches the same limit of equation (26). In the large-scale, equation (55) can be written in a compact form as

$$-3\frac{\dot\phi}{a^2\dot H}\left[\frac{1}{a}\left(\frac{a^3\dot H}{\dot\phi}\delta\phi\right)^{\cdot}\right]^{\cdot} = 0. \tag{56}$$

From this simply follows the large-scale integral form solution

$$\delta\phi = C\frac{k^2}{3}\frac{\dot\phi}{a^3\dot H}\int^t a\,dt, \tag{57}$$

where the coefficient is matched with the solutions in other gauges. (One may wonder how we, in practice, manage to make these equations into compact forms. As emphasized, if we know a solution, maybe through a suitable gauge choice, then we can derive all the other solutions even in other gauges. If we have the solution, then managing the original equation into a compact form becomes trivial.)

From equation (24) of H2 we have

$$\epsilon = \epsilon_{\rm CG} + 3H\left(\frac{1}{\mu + p} + \frac{3\gamma}{2}\frac{a^2}{k^2 - 3K}\right)^{-1}\chi_{\rm CG}. \tag{58}$$

From this relation, or from the solution in equation (57), follows all the other solutions (see Tables 1 and 2).

3.4. *Synchronous Gauge*

The SG imposes $\alpha \equiv 0$. Equations (12) and (13) leads to a third-order differential equation for $\delta\phi$:

$$\dot\kappa + 2H\kappa = \gamma(2\dot\phi\,\delta\dot\phi - V_{,\phi}\,\delta\phi), \qquad \delta\ddot\phi + 3H\,\delta\dot\phi + \left(\frac{k^2}{a^2} + V_{,\phi\phi}\right)\delta\phi = \dot\phi\kappa. \tag{59}$$

This is natural because the SG does not completely fix the gauge mode. Two of the solutions are physical whereas the remaining one solution is pure gauge construct. This is not the case in all the other gauges we introduced. Except for the SG, gauge modes are completely eliminated by each fundamental gauge-fixing condition introduced in § 2.2.

From equation (11) we have

$$\dot\chi = \varphi - H\chi \equiv \varphi_\chi, \tag{60}$$

thus

$$\chi = \int^t \varphi_{\rm ZSG}\,dt, \tag{61}$$

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where the lower bound of integration indicates the existence of a gauge mode. From this relation follows all the rest of the solutions (see Tables 1 and 2). Since $\varphi_{\rm ZSG}$ in equation (42) can be written as

$$\varphi_{\rm ZSG} = \left(C \frac{1}{a} \int^t a\,dt\right)^{\cdot} \tag{62}$$

we have

$$\chi = C \frac{1}{a} \int^t a\,dt + g\,. \tag{63}$$

where g indicates the gauge mode. From equation (11), evaluated in the SG, we can derive

$$\varphi = \frac{1}{a}(a\chi)^{\cdot} = C + gH\,. \tag{64}$$

However, since the other variables all involve $\dot{\varphi}$ in the large scale in this gauge, we cannot determine the physical modes of the variables in this method. Thus in Table 1 we present the solutions in terms of $\dot{\varphi}$, and indicate the orders of the growing modes relative to C, whereas the corresponding orders for the decaying modes are at most the same as the order of the lowest nonvanishing decaying mode of φ (check Table 4 below, where for the constant w case, the decaying mode of π is one more $[k/(aH)]^2$ order higher).

3.5. *Uniform-Curvature Gauge*

The UCG imposes $\varphi \equiv 0$. From equations (13) and (8)–(10) we can derive

$$\delta\ddot{\phi} + 3H\,\delta\dot{\phi} + \left[\frac{k^2}{a^2} + V_{,\phi\phi} + 2\frac{\dot{H}}{H}\left(3H - \frac{\ddot{H}}{\dot{H}} + 2\frac{\ddot{\phi}}{\dot{\phi}}\right)\right]\delta\phi = \frac{H}{a^3\dot{\phi}}\left[\frac{a^3\dot{\phi}^2}{H^2}\left(\frac{H}{\dot{\phi}}\,\delta\phi\right)^{\cdot}\right]^{\cdot} + \frac{k^2}{a^2}\,\delta\phi = 0\,. \tag{65}$$

Notice that this equation looks most similar to the perturbation equation based on the background metric (eq. [26]) with only additional contributions in $\delta\phi$ term. In the limit of $G \to 0$ we recover equation (26). In the small-scale, equation (65) approaches the same limit of equation (26). In the large-scale, equation (65) can be solved to give

$$\delta\phi = \frac{\dot{\phi}}{H}\left(-C + \tilde{d}\int^t \frac{H^2}{a^3\dot{\phi}^2}\,dt\right), \tag{66}$$

where the coefficient for the growing mode, C term, is matched with the solutions in other gauges; $\tilde{d}$ is a constant coefficient of the decaying mode which is $[k/(aH)]^2$ order higher than the decaying modes in the other gauge cases (see Table 1, Table 4 below, and eq. [109]); in equation (66) we do not consider the contribution from the lower bound of the integration which will contribute to the growing mode thus absorbed into C.

Compared to the equation based on the background metric, the additional terms in equation (65) modifies the mass term (assuming $V = \frac{1}{2}m^2\phi^2$) as

$$m_{\rm eff}^2 = m^2 + 2\frac{\dot{H}}{H}\left(3H - \frac{\ddot{H}}{\dot{H}} + 2\frac{\ddot{\phi}}{\dot{\phi}}\right) = m^2 + 2\frac{1}{a^3}\left(\frac{a^3\dot{H}}{H}\right)^{\cdot} \sim m^2 + 6\dot{H}\,, \tag{67}$$

where in the last step we introduced the slow-roll condition ($\ddot{\phi} \ll H\dot{\phi}$, $\dot{H} \ll H^2$). Later we will show that in both exponential and power-law expansion cases which are supported by the background scalar field dynamics, a nontrivial cancellation occurs such that equation (65) in fact becomes identical to equation (26) without the potential term. We will treat this case separately in §§ 5 and 6.

A gauge-invariant variable is introduced in Mukhanov (1988)

$$v \equiv a\left(\delta\phi - \frac{\dot{\phi}}{H}\varphi\right) = a\,\delta\phi_\varphi = -\frac{a\dot{\phi}}{H}\varphi_\Psi\,, \tag{68}$$

which becomes $a\,\delta\phi$ in the UCG and becomes $-a\dot{\phi}\varphi/H$ in the uniform-field gauge (CG). Using this variable, equation (65) becomes (in the UGC we have $v = a\,\delta\phi$)

$$v'' + \left(k^2 - \frac{z''}{z}\right)v = z^{-1}[z^2(z^{-1}v)']' + k^2 v = 0\,, \qquad z \equiv \frac{\phi'}{H}\,. \tag{69}$$

This equation looks very similar to equation (47) for u. From equations (8)–(11) one can show

$$u = -\frac{\gamma a}{2k^2}\,z\left(\frac{v}{z}\right)^{\cdot}, \qquad v = \frac{2a}{\gamma}\frac{1}{z}(zu)^{\cdot}\,. \tag{70}$$

Dimensions of these variables are

$$[v] = [z] = [\gamma^{-2}u] = [M]\,. \tag{71}$$

From equation (69) the large-scale integral form solution also simply follows. From equation (68), since v is equal to $-(a\dot{\phi}/H)\varphi$ in

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the CG, we can also derive the equation for φ in the CG as (this is only to illustrate how any gauge-invariant variable can be used in connecting variables and equations between different gauge choices)

$$\frac{H^2}{a^3\dot{\phi}^2}\left(\frac{a^3\dot{\phi}^2}{H^2}\dot{\varphi}_{CG}\right)^{\cdot}+\frac{k^2}{a^2}\varphi_{CG}=\ddot{\varphi}_{CG}+\left(3H-2\frac{\dot{H}}{H}+2\frac{\ddot{\phi}}{\dot{\phi}}\right)\dot{\varphi}_{CG}+\frac{k^2}{a^2}\varphi_{CG}=0\,. \tag{72}$$

To derive solutions from the known one in the ZSG, since

$$\varphi_\chi \equiv \varphi - H\chi\,, \tag{73}$$

we can use

$$\chi = -\frac{1}{H}\varphi_{ZSG}\,. \tag{74}$$

From this relation, or directly from equation (66), follows all the rest of the solutions in this gauge (see Tables 1 and 2). As mentioned below equation (66), the decaying mode in equation (66) is $[k/(aH)]^2$ order higher than the decaying modes of the other gauge cases, thus neglected in Table 1.

3.6. *Uniform-Density Gauge*

The UDG imposes $\epsilon \equiv 0$. From equations (12), (13), and (7) we can derive

$$\delta\ddot{\phi}+\left(5H+2\frac{\dot{H}}{H}\right)\delta\dot{\phi}+\left[\frac{k^2}{a^2}-V_{,\phi\phi}+2\frac{V_{,\phi}}{\dot{\phi}}\left(\frac{\ddot{\phi}}{\dot{\phi}}-H-\frac{\dot{H}}{H}\right)\right]\delta\phi=\frac{\dot{\phi}}{H^2a^2}\left[\frac{H^2}{a\dot{H}}\left(\frac{a^3\dot{H}}{\dot{\phi}}\delta\phi\right)^{\cdot}\right]^{\cdot}+\frac{k^2}{a^2}\delta\phi=0\,. \tag{75}$$

Later we will find that this equation is not valid for an exact de Sitter space supported by the background scalar field; in this case we need to go back to our fundamental set of equations (eqs. [8]–[13]), see § 5. In the small-scale, equation (75) does *not* approach the same limit of equation (26). The large-scale integral form solution is

$$\delta\phi = -C\frac{k^3}{3}\frac{\dot{\phi}}{a^2H\dot{H}}\left(1-\frac{H}{a}\int^t a\,dt\right), \tag{76}$$

where coefficient is matched with the solutions in other gauge.

To derive solutions from the known ones in other gauge, since

$$\epsilon_\Psi \equiv \epsilon - 3H\Psi\,, \tag{77}$$

we can use

$$\Psi = -\dot{\phi}\,\delta\phi = -\frac{1}{3H}\epsilon_{CG}\,. \tag{78}$$

From this relation, or from equation (76), follows all the other solutions (see Tables 1 and 2).

3.7. *Gauge-Invariant Variables and Comments*

Some often-used gauge-invariant (GI) variables are

$$\zeta \equiv \varphi + \frac{\epsilon}{3(\mu+p)} = \varphi + \frac{\epsilon}{3\dot{\phi}^2} = \varphi_\epsilon\,, \qquad v \equiv a\left(\delta\phi - \frac{\dot{\phi}}{H}\varphi\right) = a\,\delta\phi_\varphi\,, \qquad \varphi_\chi(\equiv -\dot{\phi}u) \equiv \varphi - H\chi\,, \qquad \epsilon_\Psi \equiv \epsilon - 3H\Psi\,. \tag{79}$$

Their values in the *large scale* are (from Table 1):

$$\zeta = C\,, \qquad v = a\,\delta\phi_\varphi = -\frac{a\dot{\phi}}{H}C\,, \qquad \varphi_\chi(= -\dot{\phi}u) = C\left(1-\frac{H}{a}\int^t a\,dt\right), \qquad \frac{\epsilon_\Psi}{\mu} = \frac{2}{3}C\left(1-\frac{H}{a}\int^t a\,dt\right)\left(\frac{k}{aH}\right)^2. \tag{80}$$

In the CG, the UEG and the UDG, both ζ and v are dominated by the φ term.

Comparing Table 1 with the ideal fluid solutions in Table 8 of H2 we can note the following: the solutions of the perturbed minimally coupled scalar field in the ZSG, the SG, and the UCG behave like the ideal fluid in the large scale; i.e., $\pi = c_s^2\epsilon$ thus $e = 0$, and solutions in the ZSG, the SG, and the UCG coincide in both cases. In the large scale, only α and π/μ in the CG, the UEG, and the UDG are different. The variables α and π/μ in these gauges are $(k/aH)^2$ order higher than in the other gauges; i.e., the dominating mode has canceled. Of course, $\delta\phi$ does not exist in an ideal fluid case.

From Table 1 we can note that for the ZSG, the UCG and the SG we have

$$\epsilon = -3H\dot{\phi}\,\delta\phi\,. \tag{81}$$

Only in the slow-roll approximation of the background scalar field we have $\epsilon = V_{,\phi}\delta\phi$ which can be obtained from naively neglecting the first two terms in equation (7).

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In the *small scale* we have (from Table 2):

$$\zeta = \frac{i\sqrt{k}}{3\sqrt{2}a^2\dot{\phi}}(c_1 e^{ik\eta} + c_2 e^{-ik\eta}) , \qquad v = a\,\delta\phi_\varphi = \frac{1}{\sqrt{2k}}(c_1 e^{ik\eta} + c_2 e^{-ik\eta}) ,$$

$$\varphi_\chi(= -\dot{\phi}u) = \frac{i\gamma\dot{\phi}}{(2k)^{3/2}}(c_1 e^{ik\eta} - c_2 e^{-ik\eta}) , \qquad \frac{\epsilon_\Psi}{\mu} = \frac{i\gamma\sqrt{k}\dot{\phi}}{3\sqrt{2}a^2H^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta}) . \tag{82}$$

For the ZSG, the UEG, the SG, and the UCG, v is dominated by the $\delta\phi$ term. Also, except for the CG and the UDG, we have

$$\delta\phi = \frac{1}{\sqrt{2k}a}[c_1(k)e^{ik\eta} + c_2(k)e^{-ik\eta}] . \tag{83}$$

We showed that in the small scale the perturbed equations for $\delta\ddot{\phi}$, except for the CG and the UDG, *approach* the equation based on the background metric (eq. [26]) without the potential term

$$\delta\ddot{\phi} + 3H\,\delta\dot{\phi} + a^{-2}k^2\,\delta\phi = 0 . \tag{84}$$

Except for the CG and the UDG (also the uniform-pressure gauge), all fluid quantities (i.e., ϵ, π, $\delta\phi$ and equivalently Ψ) behave the same way independent of the gauge choice. Except for UDC, we have

$$\frac{\epsilon}{\mu} = \frac{\pi}{\mu} = \frac{i\gamma\sqrt{k}\dot{\phi}}{3\sqrt{2}a^2H^2}(c_1 e^{ik\eta} - c_2 e^{-ik\eta}) . \tag{85}$$

Except for the UDG, ζ is dominated by ϵ term.

We can check the different normalization conditions for different variables

$$v\dot{v}^* - \dot{v}v^* = ia^{-1} , \qquad u\dot{u}^* - \dot{u}u^* = ia^{-1}\left(\frac{\gamma}{2k}\right)^2 , \quad \text{etc.} \tag{86}$$

Sasaki (986) used a related variable

$$Q \equiv -a^{-1}\frac{2k}{\gamma\dot{\phi}}\varphi_\chi = a^{-1}\frac{2k}{\gamma}u . \tag{87}$$

According to our normalization of $\delta\phi$ in the small scale, which gives equation (86), Q should follow

$$Q\dot{Q}^* - \dot{Q}Q^* = ia^{-3} . \tag{88}$$

Using the analogy of the derived equation for Q, which is essentially our equation (38) and equivalently equation (47), Sasaki quantized Q itself, which by the correct normalization taken (eqs. [87]–[88]) the derived result is consistent with our general presentation of the whole solutions. (This was incorrectly pointed out in Deruelle, Gundlach, & Polarski 1992; they used a variable $Y = -2\varphi_\chi/(\gamma\dot{\phi}) = 2u/\gamma$, which leads to $Y\dot{Y}^* - \dot{Y}Y^* = ia^{-1}k^{-2}$, which is, of course, consistent with eqs. [87]–[88].)

3.8. *Large-Scale Conserved Variables*

We note that ζ is a GI variable which remains constant in the large scale with its dominating decaying mode vanishing. From equations (22) and (27) of H1 we can show (for general K, Λ, and σ)

$$\dot{\zeta} = \frac{1}{\mu + p}\left(\frac{k^2}{3a^2}\Psi_\chi - He\right) , \tag{89}$$

where $\Psi_\chi \equiv \Psi + (\mu + p)\chi$ and e is defined below equation (40). For the scalar field case (using eqs. [7] and [41]) we have (for general K, Λ, and σ)

$$\dot{\zeta} = -\frac{k^2}{a^2}\left[-\frac{1}{3\dot{\phi}}\delta\phi_\chi + \frac{2}{\gamma}(1 - c_s^2)H\varphi_\chi\right] , \tag{90}$$

where $\delta\phi_\chi \equiv \delta\phi - \dot{\phi}\chi$ and $\varphi_\chi \equiv \varphi - H\chi$. (We note that these GI combinations of the variables can be uniquely determined from the gauge-transformation property of each variable presented in § 2.2 of H1.) Thus, we see that in the large scale, ζ is conserved for general background evolution.

From Table 1 we can also note that except for the ZSG, *the perturbed curvature* (or potential) *variable φ in all the other gauge choices remain constant* as

$$\varphi = C . \tag{91}$$

For the ZSG we have equation (42). Of course, the UCG is an exception with $\varphi = 0$. In the SG we have a contribution from the gauge mode which can be removed by proper choice of the initial spatial hypersurface. This fact can be stated using GI combinations of the variables (e.g., $\varphi_{\mathrm{UEG}} = \varphi_\kappa$, etc.) as

$$\varphi_{\delta\phi} = \varphi_\kappa = \varphi_\alpha = \varphi_\epsilon = \varphi_\pi = C , \tag{92}$$

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where

$$\varphi_{\delta\phi} \equiv \varphi - \frac{H}{\dot\phi}\,\delta\phi \equiv -\frac{H}{\dot\phi}\,\delta\phi_\varphi\,, \qquad \varphi_\kappa \equiv \varphi + \frac{H}{3\dot H - k^2/a^2}\,\kappa \equiv \frac{H}{3\dot H - k^2/a^2}\,\kappa_\varphi\,, \qquad \varphi_\alpha = \varphi - H\int^t \alpha\,dt \equiv -H\int^t \alpha_\varphi\,dt\,,$$

$$\varphi_\epsilon \equiv \varphi + \frac{\epsilon}{3(\mu + p)} \equiv \frac{\epsilon_\varphi}{3(\mu + p)}\,, \qquad \varphi_\pi \equiv \varphi + \frac{\pi}{3c_s^2(\mu + p)} \equiv \frac{\pi_\varphi}{3c_s^2(\mu + p)}\,. \tag{93}$$

The variable φ_α is not GI (this is generally the case for the combination with subindex α); it has a gauge-mode contribution from the lower bound of the integration. All the other combinations are GI (see § 3.2.1 of H1). The result of φ_π follows from π_φ presented in Table 1. The variable φ_ϵ is in fact ζ, and $\varphi_{\delta\phi}$ is also often used in the literature (Sasaki 1986; Mukhanov 1988). Notice that all the variables in the right-hand side of equation (93) are variables in the UCG; $(H/\dot\phi)\delta\phi_\varphi$, etc., are also conserved with their dominating decaying mode vanishing, where again the exception is the case for $\varphi_\chi \equiv \varphi - H\chi \equiv -H\chi_\varphi$.

In the ZSG, φ is not conserved for the generally changing background equations of state (eq. [42]):

$$\left(1 - \frac{H}{a}\int^t a\,dt\right)^{-1}\varphi_\chi = C\,, \tag{94}$$

which becomes, for constant w,

$$\left[1 + \frac{2}{3(1 + w)}\right]\varphi_\chi = C\,. \tag{95}$$

This result is often used in deriving the inflationary spectrum previously. We can note that actually the UCG provides most simple analysis available in treating the scalar field perturbation problem; this is studied in detail in Hwang (1993b). The variable φ_χ is connected to the density perturbation in the CG through equation (51), and thus has direct Newtonian correspondence; in this sense, despite its dependence on background equation of state, the behavior of φ_χ is conveniently important for characterizing the Newtonian potential (and density) fluctuations. Thus the behavior of δ in the CG can be written as (for $K = 0 = \Lambda$)

$$C = \left(1 - \frac{H}{a}\int^t a\,dt\right)^{-1}\frac{3}{2}\left(\frac{aH}{k}\right)^2\delta_\Psi = \left[1 + \frac{2}{3(1 + w)}\right]\frac{3}{2}\left(\frac{aH}{k}\right)^2\delta_\Psi\,, \tag{96}$$

which was also derived directly in equation (52).

From Table 1, we can find one other conserved combination (with its dominating decaying mode vanishing), which is

$$\kappa_\epsilon \equiv \kappa - \frac{3\dot H - k^2/a^2}{3H(\mu + p)}\,\epsilon \equiv -\frac{3\dot H - k^2/a^2}{3H(\mu + p)}\,\epsilon_\kappa \simeq C\,\frac{k^2}{a^2 H}\,, \tag{97}$$

where the last step hold for the large scale (Table 1). Thus $a^2 H\kappa_\epsilon$, or equivalently $a^2\epsilon_\kappa$, remains conserved in the large scale.

4. EXPONENTIAL POTENTIAL CASE

For $w =$ constant, from equations (15) and (16), thus *assuming* $K = 0 = \Lambda$, we have

$$a \propto t^{2/[3(1+w)]}\,, \qquad \phi = \sqrt{\frac{4}{3\gamma(1 + w)}}\ln t\,, \qquad V = \frac{2(1 - w)}{3\gamma(1 + w)^2}\,e^{-\sqrt{3\gamma(1+w)}\phi}\,. \tag{98}$$

Inflationary models based on this case are often called the exponential potential or the power-law expansion inflation (Lucchin & Matarrese 1985).

A particular case of $V = 0$ corresponds to a ultrarelativistic stiff fluid with $w = 1$. In this case, from equations (5) and (6) we can derive

$$\dot\phi = \sqrt{\frac{2}{3\gamma}}\,t^{-1} \equiv ca^{-3}\,, \qquad a = \left(\frac{3\gamma}{2}\right)^{1/6}(ct)^{1/3}\,, \qquad \mu = p = \frac{\dot\phi^2}{2} = (3\gamma t^2)^{-1} = \frac{c^2}{2}\,a^{-6}\,. \tag{99}$$

The solutions of this case can be included in our general treatment in equation (98).

4.1. *Exact Solutions*

In this case, equation (49) can be written as a Bessell function. Following Bardeen (1980), we introduce a set of variables as:

$$f(x) \equiv x^{\beta - 2}\delta_{CG}\,, \qquad x \equiv k\eta = \beta\,\frac{k}{aH}\,, \quad (a \propto \eta^\beta)\,, \qquad \beta = \frac{2}{3w + 1} = -\frac{V + \dot\phi^2/2}{V - \dot\phi^2}\,. \tag{100}$$

Equation (49) gives

$$x^2\partial_x^2 f + 2x\partial_x f + [x^2 - \beta(\beta + 1)]f = 0\,. \tag{101}$$

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The solutions can be written using the spherical Bessel functions as

$$f(x) = \hat{a} j_\beta(x) + \hat{b} n_\beta(x) , \tag{102}$$

where $\hat{a}$ and $\hat{b}$ are constant (in time) coefficients which generally depend on k. Notice that the arguments of the Bessel functions are just x. This is in contrast with a w = constant ideal fluid case where it was $(w^{1/2})x$ (compare with eq. [5.5] of Bardeen 1980). Thus in contrast to the ideal fluid case, the $w = 0$ ($\beta = 2$) case can be generally treated by solution in equation (102); in the ideal fluid case, for $w = 0$ we have a simpler analytic form of exact solution (see eq. [29] of H2).

Following the procedure presented in previous section we can derive the complete set of exact solutions in an implicit form using f. These solutions are presented in Table 3. We can compare the functional forms of the solutions in Table 3 with the exact solutions for the ideal fluid case presented in Table 1 of H2. The solutions in these two cases are the same except for α's in the CG, the UEG, and the UDG, and π/μ's in every gauge. Still, we would like to remind that f itself is different from the one in the ideal fluid case (see § 3.4 of H1). Solutions in the SG case have also been derived in Ratra (1992); in that paper, explicit integrations of the SG solutions in Table 3 were done in terms of the Lommel functions, where the gauge modes are traced separately. In our approach, the gauge modes in the SG are explicitly distinguished as the lower bound of the integration of the $\int^x f\, dx$ terms.

Using the asymptotic expansions of f in equation (102) we can derive the complete large- and small-scale asymptotic solutions in every gauge.

4.2. *Large-Scale Solution*

In the large scale, $f(x)$ can be expanded as

$$f(x) = c x^\beta \left[1 - \frac{x^2}{2(2\beta + 3)} + \cdots \right] + d x^{-\beta-1} \left[1 + \frac{x^2}{2(2\beta - 1)} + \frac{x^4}{8(2\beta - 3)(2\beta - 1)} + \cdots \right] . \tag{103}$$

The constant coefficients c and d generally depend on k and indicate the growing and the decaying modes, respectively:

$$c = \hat{a} \frac{2^\beta \Gamma(\beta + 1)}{\Gamma(2\beta + 2)} , \qquad d = -\hat{b} \frac{\Gamma(2\beta + 1)}{2^\beta \Gamma(\beta + 1)} . \tag{104}$$

We expand the growing mode to second order and the decaying mode to third order because, especially in the SG case, some lower order terms cancel out. Apparently, we should separately consider certain β's leading to singularities in f. For those β's the solutions may end up with additional logarithmic dependences. The complete nonvanishing large-scale dominating modes for the constant w case are presented in Table 4. We note that compared to Table 1, although Table 4 is restricted to the constant w case, it contains the lowest order nonvanishing mode for each solution. Table 1 contains only the dominating large-scale modes. By defining

$$c \equiv \frac{2(\beta + 1)}{3\beta^2(2\beta + 1)} C , \tag{105}$$

we can check that the solutions derived in Table 4 correspond to the constant w limit of the solutions in Table 1. In § 3.1 we showed that C is constant, independent of changing w for general potential. The decaying modes of Tables 1 and 4 are related by

$$C \frac{H}{a} \int a\, dt \bigg|_{w = \text{constant}} \equiv -\frac{3}{2} \beta^2 d x^{-2\beta-1} , \tag{106}$$

where we do not consider the contribution of the upper bound of the integration to the growing mode. By defining

$$a \equiv a_1 x^\beta ; \quad \text{thus} \quad t = \frac{a_1}{k(1 + \beta)} x^{\beta+1} , \tag{107}$$

we have (no upper bound considered)

$$C \int a\, dt = -\frac{3\beta a_1^2}{2k} d , \tag{108}$$

and the coefficient $\tilde{d}$ in equation (66) can be related to d as

$$\tilde{d} = -\frac{3\beta}{\gamma k a_1^2} d . \tag{109}$$

The gauge modes in the SG are related by (see eqs. [61] and [63])

$$\int \varphi_\chi\, dt \bigg|_{w = \text{constant}} \equiv g , \qquad \int f\, dx \equiv \tilde{g} , \qquad H\tilde{g} \equiv x^{-\beta-1} g , \tag{110}$$

where we do not consider the contributions from the upper bounds of the integrations which will contribute to physical modes.

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TABLE 3

EXACT SOLUTIONS FOR w = CONSTANT CASE

Variables/Gauges	$\Psi = 0 = \delta\phi$ (Comoving)	$\chi = 0$ (Zero Shear)	$\kappa = 0$ (Uniform Expansion)
$\gamma\Psi/H$	0	$3\beta x^{-\beta+1}\partial_x f$	$A^{-1}\dfrac{1}{1+\beta}x^{-\beta+3}\partial_x f$
$H\,\delta\phi/\dot{\phi}$	0	$-\dfrac{3\beta^2}{2(1+\beta)}x^{-\beta+1}\partial_x f$	$-A^{-1}\dfrac{\beta}{2(1+\beta)^2}x^{-\beta+3}\partial_x f$
$H\chi$	$\dfrac{3\beta^2}{2(1+\beta)}x^{-\beta+1}\partial_x f$	0	$A^{-1}\dfrac{3\beta^2}{2(1+\beta)}x^{-\beta+1}\partial_x f$
κ/H	$\dfrac{3}{2(1+\beta)}x^{-\beta+3}\partial_x f$	$-\dfrac{9\beta}{2}x^{-\beta+1}\partial_x f$	0
α	$-\dfrac{3\beta}{2(1+\beta)}x^{-\beta+2}f$	$-\dfrac{3\beta^2}{2}x^{-\beta}f$	$-A^{-2}\dfrac{1}{(1+\beta)^2}x^{-\beta+3}\partial_x f - A^{-1}\dfrac{2\beta}{1+\beta}x^{-\beta+2}f$
φ	$\dfrac{3\beta^2}{2(1+\beta)}x^{-\beta+1}\partial_x f + \dfrac{3}{2}\beta^2 x^{-\beta}f$	$\dfrac{3\beta^2}{2}x^{-\beta}f$	$A^{-1}\dfrac{3\beta^2}{2(1+\beta)}x^{-\beta+1}\partial_x f + \dfrac{3\beta^2}{2}x^{-\beta}f$
ϵ/μ	$x^{-\beta+2}f$	$3\beta x^{-\beta+1}\partial_x f + x^{-\beta+2}f$	$A^{-1}\dfrac{1}{1+\beta}x^{-\beta+3}\partial_x f + x^{-\beta+2}f$
π/μ	$x^{-\beta+2}f$	$(2-\beta)x^{-\beta+1}\partial_x f + x^{-\beta+2}f$	$A^{-1}\dfrac{2-\beta}{3\beta(1+\beta)}x^{-\beta+3}\partial_x f + x^{-\beta+2}f$

Variables/Gauges	$\alpha = 0$ (Synchronous)	$\varphi = 0$ (Uniform Curvature)	$\epsilon = 0$ (Uniform Density)
$\gamma\Psi/H$	$3\beta x^{-\beta+1}\partial_x f - 3\beta^2(1+\beta)x^{-\beta-1}\int^x f\,dx$	$3\beta x^{-\beta+1}\partial_x f + 3\beta(1+\beta)x^{-\beta}f$	$-x^{-\beta+2}f$
$H\delta\phi/\dot{\phi}$	$-\dfrac{3\beta^2}{2(1+\beta)}x^{-\beta+1}\partial_x f + \dfrac{3\beta^3}{2}x^{-\beta-1}\int^x f\,dx$	$-\dfrac{3\beta^2}{2(1+\beta)}x^{-\beta+1}\partial_x f - \dfrac{3\beta^2}{2}x^{-\beta}f$	$\dfrac{\beta}{2(1+\beta)}x^{-\beta+2}f$
$H\chi$	$\dfrac{3\beta^3}{2}x^{-\beta-1}\int^x f\,dx$	$-\dfrac{3\beta^2}{2}x^{-\beta}f$	$\dfrac{3\beta^2}{2(1+\beta)}x^{-\beta+1}\partial_x f + \dfrac{\beta}{2(1+\beta)}x^{-\beta+2}f$
κ/H	$-\dfrac{9\beta}{2}x^{-\beta+1}\partial_x f + \dfrac{9\beta^2(1+\beta)}{2}Ax^{-\beta-1}\int^x f\,dx$	$-\dfrac{9\beta}{2}x^{-\beta+1}\partial_x f - \dfrac{9\beta(1+\beta)}{2}Ax^{-\beta}f$	$\dfrac{3}{2(1+\beta)}x^{-\beta+3}\partial_x f + \dfrac{3}{2}Ax^{-\beta+2}f$
α	0	$-\dfrac{3\beta}{2}x^{-\beta+1}\partial_x f - \dfrac{3\beta(1+\beta)}{2}x^{-\beta}f$	$\dfrac{1}{2(1+\beta)}x^{-\beta+3}\partial_x f + \dfrac{3(1-\beta)}{2(1+\beta)}x^{-\beta+2}f$
φ	$\dfrac{3\beta^2}{2}x^{-\beta}f + \dfrac{3\beta^3}{2}x^{-\beta-1}\int^x f\,dx$	0	$\dfrac{3\beta^2}{2(1+\beta)}x^{-\beta+1}\partial_x f + \dfrac{3\beta^2}{2}Ax^{-\beta}f$
ϵ/μ	$3\beta x^{-\beta+1}\partial_x f + x^{-\beta+2}f - 3\beta^2(1+\beta)x^{-\beta-1}\int^x f\,dx$	$3\beta x^{-\beta+1}\partial_x f + 3\beta(1+\beta)Ax^{-\beta}f$	0
π/μ	$(2-\beta)x^{-\beta+1}\partial_x f + x^{-\beta+2}f - \beta(1+\beta)(2-\beta)x^{-\beta-1}\int^x f\,dx$	$(2-\beta)x^{-\beta+1}\partial_x f + [x^2 - (\beta+1)(\beta-2)]x^{-\beta}f$	$\dfrac{2(2\beta-1)}{3\beta}x^{-\beta+2}f$

NOTES.—$f(x)$ is given in eq. (102); $A \equiv 1 + x^2/[3\beta(1+\beta)]$; $\partial_x \equiv k^{-1}\partial_\eta$. In the SG, the lower bound of integration causes the behavior of the gauge mode.

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TABLE 4

LARGE-SCALE SOLUTION FOR w = CONSTANT

Variables/Gauges	$\Psi = 0 = \delta\phi$ (Comoving)	$\chi = 0$ (Zero Shear)	$\kappa = 0$ (Uniform Expansion)
$\gamma\Psi/H$	0	$C\frac{2(\beta+1)}{2\beta+1} - d3\beta(\beta+1)x^{-2\beta-1}$	$C\frac{2}{3\beta(2\beta+1)}x^2 - dx^{-2\beta+1}$
$H\,\delta\phi/\dot\phi$	0	$-C\frac{\beta}{2\beta+1} + d\frac{3}{2}\beta^2 x^{-2\beta-1}$	$-C\frac{1}{3(\beta+1)(2\beta+1)}x^2 + d\frac{\beta}{2(\beta+1)}x^{-2\beta+1}$
$H\chi$	$C\frac{\beta}{2\beta+1} - d\frac{3}{2}\beta^2 x^{-2\beta-1}$	0	$C\frac{\beta}{2\beta+1} - d\frac{3}{2}\beta^2 x^{-2\beta-1}$
κ/H	$C\frac{1}{\beta(2\beta+1)}x^2 - d\frac{3}{2}x^{-2\beta+1}$	$-C\frac{3(\beta+1)}{2\beta+1} + d\frac{9}{2}\beta(\beta+1)x^{-2\beta-1}$	0
α	$-C\frac{1}{\beta(2\beta+1)}x^2 - d\frac{3\beta}{2(\beta+1)}x^{-2\beta+1}$	$-C\frac{\beta+1}{2\beta+1} - d\frac{3}{2}\beta^2 x^{-2\beta-1}$	$-C\frac{2(2\beta+3)}{3\beta(\beta+1)(2\beta+1)}x^2 - d\frac{2\beta-1}{\beta+1}x^{-2\beta+1}$
φ	$C + d\frac{3\beta^2}{2(\beta+1)(2\beta-1)}x^{-2\beta+1}$	$C\frac{\beta+1}{2\beta+1} + d\frac{3}{2}\beta^2 x^{-2\beta-1}$	$C + d\frac{5\beta-1}{2(\beta+1)(2\beta-1)}x^{-2\beta+1}$
ϵ/μ	$C\frac{2(\beta+1)}{3\beta^2(2\beta+1)}x^2 + dx^{-2\beta+1}$	$C\frac{2(\beta+1)}{2\beta+1} - d3\beta(\beta+1)x^{-2\beta-1}$	$C\frac{2}{3\beta^2}x^2 + d\frac{2}{3(2\beta-1)}x^{-2\beta+3}$
π/μ	$C\frac{2(\beta+1)}{3\beta^2(2\beta+1)}x^2 + dx^{-2\beta+1}$	$-C\frac{2(\beta+1)(\beta-2)}{3\beta(2\beta+1)} + d(\beta+1)(\beta-2)x^{-2\beta-1}$	$C\frac{2(2\beta+5)}{9\beta^2(2\beta+1)}x^2 + d\frac{2(2\beta-1)}{3\beta}x^{-2\beta+1}$

Variables/Gauges	$\alpha = 0$ (Synchronous)	$\varphi = 0$ (Uniform Curvature)	$\epsilon = 0$ (Uniform Density)
$\gamma\Psi/H$	$-C\frac{2(\beta+1)}{\beta(2\beta+1)(\beta+3)}x^2 + d\frac{3\beta}{\beta-2}x^{-2\beta+1} - g\frac{2(\beta+1)}{\beta}x^{-\beta-1}$	$C\frac{2(\beta+1)}{\beta} + d\frac{3\beta}{2\beta-1}x^{-2\beta+1}$	$-C\frac{2(\beta+1)}{3\beta^2(2\beta+1)}x^2 - dx^{-2\beta+1}$
$H\,\delta\phi/\dot\phi$	$C\frac{1}{(2\beta+1)(\beta+3)}x^2 - d\frac{3}{2(\beta+1)(\beta-2)}x^{-2\beta+1} + gx^{-\beta-1}$	$-C - d\frac{3\beta^2}{2(\beta+1)(2\beta-1)}x^{-2\beta+1}$	$C\frac{1}{3\beta(2\beta+1)}x^2 + d\frac{\beta}{2(\beta+1)}x^{-2\beta+1}$
$H\chi$	$C\frac{\beta}{2\beta+1} - d\frac{3}{2}\beta^2 x^{-2\beta-1} + gx^{-\beta-1}$	$-C\frac{\beta+1}{2\beta+1} - d\frac{3}{2}\beta^2 x^{-2\beta-1}$	$C\frac{\beta}{2\beta+1} - d\frac{3}{2}\beta^2 x^{-2\beta-1}$
κ/H	$C\frac{2(2\beta+3)}{\beta^2(\beta+3)(2\beta+1)}x^2 - d\frac{3(2\beta-1)}{\beta-2}x^{-2\beta+1} + g\frac{3(\beta+1)}{\beta}x^{-\beta-1}$	$-C\frac{3(\beta+1)}{\beta} + d\frac{9(5\beta-3)}{2(2\beta-1)}x^{-2\beta+1}$	$C\frac{1}{\beta^2}x^2 + d\frac{1}{2\beta-1}x^{-2\beta+3}$
α	0	$-C\frac{\beta+1}{\beta} - d\frac{3\beta}{2(2\beta-1)}x^{-2\beta+1}$	$-C\frac{2\beta-3}{3\beta^2(2\beta+1)}x^2 - d\frac{2\beta-1}{\beta+1}x^{-2\beta+1}$
φ	$C - d\frac{3\beta^2}{2(2\beta-1)(\beta-2)}x^{-2\beta+1} + gx^{-\beta-1}$	0	$C + d\frac{\beta^2}{2\beta-1}x^{-2\beta+1}$
ϵ/μ	$-C\frac{2(\beta+1)(2\beta-3)}{3\beta^2(2\beta+1)(\beta+3)}x^2 + d\frac{2(2\beta-1)}{\beta-2}x^{-2\beta+1} - g\frac{2(\beta+1)}{\beta}x^{-\beta-1}$	$C\frac{2(\beta+1)}{\beta} + d\frac{5\beta-1}{2\beta-1}x^{-2\beta+1}$	0
π/μ	$C\frac{2(\beta+1)}{3\beta^2(\beta+3)}x^2 - d\frac{1}{(2\beta-1)(\beta-4)}x^{-2\beta+3} + g\frac{2(\beta+1)(\beta-2)}{3\beta^2}x^{-\beta-1}$	$-C\frac{2(\beta+1)(\beta-2)}{3\beta^2} + d\frac{\beta+1}{2\beta-1}x^{-2\beta+1}$	$C\frac{4(\beta+1)(2\beta-1)}{9\beta^3(2\beta+1)}x^2 + d\frac{2(2\beta-1)}{3\beta}x^{-2\beta+1}$

NOTES.—Compared to Table 1, the decaying modes are presented to lowest nonvanishing order. In the cases where the coefficient diverge, e.g., for $\beta = -1, \frac{1}{2}, 2, -3, 4$, etc., some logarithmic corrections may arise.

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4.3. *Small-Scale Solutions*

In the small scale, to match the solutions in Table 2, we rewrite the exact solution in equation (102) in the following way:

$$f(x) = \sqrt{\frac{\pi}{2x}}\,[\hat{c}_1 e^{i(1+\beta)\pi/2} H^{(1)}_{\beta+1/2}(x) + \hat{c}_2 e^{-i(1+\beta)\pi/2} H^{(2)}_{\beta+1/2}(x)]\,, \tag{111}$$

where

$$\hat{c}_1 e^{i(1+\beta)\pi/2} = \tfrac{1}{2}(\hat{a} - i\hat{b})\,, \qquad \hat{c}_2 e^{-i(1+\beta)\pi/2} = \tfrac{1}{2}(\hat{a} + i\hat{b})\,. \tag{112}$$

Thus, in the small scale ($x \gg 1$) we have

$$f(x) = \frac{1}{x}(\hat{c}_1 e^{ix} + \hat{c}_2 e^{-ix})\,. \tag{113}$$

To match the solutions in Table 2 we let

$$\hat{c}_1 e^{ix} + \hat{c}_2 e^{-ix} \equiv iB[c_1(k)e^{ix} - c_2(k)e^{-ix}] = \hat{a}\cos\left(x - \frac{1+\beta}{2}\pi\right) + \hat{b}\sin\left(x - \frac{1+\beta}{2}\pi\right), \tag{114}$$

where $c_1(k)$ and $c_2(k)$ follow equation (29). Using these variables, the complete solutions are presented in Table 5. With

$$B \equiv \frac{1}{3\beta^{3/2}}\sqrt{\frac{\gamma(1+\beta)}{k}}\,\frac{1}{a_1}\,, \tag{115}$$

we can show that in the constant w limit the solutions in Table 2 become the same as the solutions in Table 5.

We can match the large-scale and the small-scale coefficients. From equations (104), (105), and (114) we can show that

$$C = i\,\frac{3\cdot 2^{\beta-1}\beta^2\Gamma(\beta+1)}{(\beta+1)\Gamma(2\beta+1)}\,B[c_1(k)e^{i(1+\beta)\pi/2} - c_2(k)e^{-i(1+\beta)\pi/2}]\,, \qquad d = \frac{\Gamma(2\beta+1)}{2^\beta\Gamma(\beta+1)}\,B[c_1(k)e^{i(1+\beta)\pi/2} + c_2(k)e^{-i(1+\beta)\pi/2}]\,. \tag{116}$$

Thus in adiabatic vacuum we have

$$d = i\,\frac{(\beta+1)\Gamma^2(2\beta+1)}{3\times 2^{2\beta-1}\Gamma^2(\beta+1)}\,C\,. \tag{117}$$

5. DE SITTER BACKGROUND CASE

In this section we consider an *exact de Sitter background* space *realized by* a special choice of background scalar field. *Assuming* $K = 0 = \Lambda$, since we have $H =$ constant, equations (5) and (6) imply

$$\dot{H} = 0\,, \qquad \dot{\phi} = 0\,, \qquad V_{,\phi} = 0\,, \qquad \mu = -p = V = \text{constant}\,. \tag{118}$$

Thus considering only the background, it is equivalent to introducing a cosmological constant with $\Lambda = \gamma\mu = \gamma V$ which is constant both in space and time. In fact, *from* the fact that μ, p, and ϕ are constant both in space and time follow that all the fluid quantities, ϵ, π, and Ψ thus $\delta\phi$ are GI in de Sitter space. This can be directly checked from equations (17)–(19) of H1. In this sense, de Sitter space is a degenerate case. A similar case occurs in the ideal fluid (H2). (In fact, all the perturbed fluid quantities, ϵ, π, and Ψ, vanish; see eq. [7]. However, $\delta\phi$ does not necessary vanish.) In this special case we need to go back to our original set of equations, (8)–(13) (see remark below eq. [75]), which give

$$\kappa = -3\dot{\varphi} + 3H\alpha + \frac{k^2}{a^2}\chi\,, \qquad -H\kappa + \frac{k^2}{a^2}\varphi = 0\,, \qquad \kappa - \frac{k^2}{a^2}\chi = 0\,,$$

$$\alpha + \varphi = \dot{\chi} + H\chi\,, \qquad \dot{\kappa} + 2H\kappa = \frac{k^2}{a^2}\alpha\,, \qquad \delta\ddot{\phi} + 3H\,\delta\dot{\phi} + \frac{k^2}{a^2}\delta\phi = 0\,. \tag{119}$$

Notice that the first five equations for the metric perturbations are completely free of $\delta\phi$, and also the last equation for $\delta\phi$ does not depend on the perturbed metric perturbations and can be identified with equation (26) without potential term. Thus the general solution derived in § 2.4 with $m = 0$, thus $v = 3/2$ applies in this case. (We note that the perturbed fluid quantities in eq. [7] vanish, thus the original form of eq. [27] of H1 vanishes identically. However, notice that the fluid quantities divided by $\dot{\phi}$ and also eq. [27] of H1 divided by $\dot{\phi}$ do not vanish identically and the second one leads to the last of eq. [119].)

The metric perturbation part of equation (119) can be presented as

$$\kappa = \frac{k^2}{a^2}\frac{\varphi}{H}\,, \qquad \chi = \frac{\varphi}{H}\,, \qquad \alpha = \left(\frac{\varphi}{H}\right)^{\cdot}\,, \tag{120}$$

where φ can be an arbitrary function of time. Thus the UCG, the UEG, and the ZSG all imply the complete vanishing of the metric perturbation variables; notice that ϵ, Ψ, and $\delta\phi$ are GI. However, the SG, thus fixing $\alpha = 0$, still leave room for the gauge mode as

$$\varphi \equiv \hat{g}H\,, \qquad \chi = \hat{g}\,, \qquad \kappa = \hat{g}\frac{k^2}{a^2}\,. \tag{121}$$

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TABLE 5

SMALL-SCALE SOLUTIONS

Variables/Gauges	$\Psi = 0 = \delta\phi$ (Comoving)	$\chi = 0$ (Zero Shear)	$\kappa = 0$ (Uniform Expansion)	$\alpha = 0$ (Synchronous)
$\gamma\Psi/H$	0	$-3\beta x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$-3\beta x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$-3\beta x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$
$H\,\delta\phi/\dot{\phi}$	0	$\frac{3\beta^2}{2(\beta+1)} x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{3\beta^2}{2(\beta+1)} x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{3\beta^2}{2(\beta+1)} x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$
$H\chi$	$-\frac{3\beta^2}{2(\beta+1)} x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	0	$-\frac{9\beta^3}{2} x^{-\beta-2}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{3\beta^3}{2} x^{-\beta-2}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$
κ/H	$-\frac{3}{2(\beta+1)} x^{-\beta+2}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{9\beta}{2} x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	0	$6\beta x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$
α	$-\frac{3\beta}{2(\beta+1)} x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$-\frac{3\beta^2}{2} x^{-\beta-1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$-6\beta^2 x^{-\beta-1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	0
φ	$-\frac{3\beta^2}{2(\beta+1)} x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{3\beta^2}{2} x^{-\beta-1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{3\beta^2}{2} x^{-\beta-1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{3\beta^2}{2} x^{-\beta-1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
ϵ/μ	$x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
π/μ	$x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$

Variables/Gauges	$\varphi = 0$ (Uniform Curvature)	$\epsilon = 0$ (Uniform Density)
$\gamma\Psi/H$	$-3\beta x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$-x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
$H\,\delta\phi/\dot{\phi}$	$\frac{3\beta^2}{2(\beta+1)} x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$\frac{\beta}{2(\beta+1)} x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
$H\chi$	$-\frac{3\beta^2}{2} x^{-\beta-1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{\beta}{2(\beta+1)} x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
κ/H	$-\frac{3}{2} x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{1}{2\beta(\beta+1)} x^{-\beta+3}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
α	$\frac{3\beta}{2} x^{-\beta}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$	$-\frac{1}{2(\beta+1)} x^{-\beta+2}B(c_1 e^{ik\eta} + c_2 e^{-ik\eta})$
φ	0	$\frac{\beta}{2(\beta+1)} x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$
ϵ/μ	$x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	0
π/μ	$x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$	$\frac{2(2\beta-1)}{3\beta} x^{-\beta+1}iB(c_1 e^{ik\eta} - c_2 e^{-ik\eta})$

NOTES.—We neglect the gauge modes in the SG. The coefficients c_1 and c_2 indicate $c_1(k)$ and $c_2(k)$, respectively. If we impose quantization condition these coefficients follow eq. [29].

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We note again that in this section we considered a pure de Sitter background which is a degenerate case. This should be treated separately from the case where w is very close to -1, which can be considered as representing ordinary inflationary stages and is generally treated in previous sections.

6. UNIFORM-CURVATURE GAUGE

In some (inflationary) expansion stages the perturbed scalar field equation in the UCG (eq. [65]) simplifies to *the form* of the perturbed minimally coupled scalar field equation based on a fixed uperturbed metric (eq. [26]).

Since the exponential expansion is characterized by $H =$ constant, equation (65) simplifies to equation (26). This applies to most of the gauge choices for a near de Sitter space case. This is also true independent of any gauge choice for an exact de Sitter space, § 5.

In the power-law expansion stage *supported by the MSF* (eq. [98]), one can show that an interesting cancellation occurs such that

$$V_{,\phi\phi} + 2\frac{\dot{H}}{H}\left(3H - \frac{\dot{H}}{H} + 2\frac{\ddot{\phi}}{\dot{\phi}}\right) = 0\,, \tag{122}$$

thus equation (65) again simplifies (in exact manner) to equation (26). Thus, following the same procedure as in § 2.4 we can show that the mode function solution remains the same as in equation (32); notice, however, that both the exponential expansion in equation (27) and the power-law expansion in equation (31) treated in § 2.4 (where we neglect the perturbed metric contributions) were *not supported by the MSF*. Thus, deriving the vacuum quantum fluctuations become trivial in both near-exponential or power-law expansion backgrounds; these were in fact derived in equations (35) and (36), respectively.

Using the known solutions in the UCG we can derive all the other solutions in any other gauge; the following translations are generally valid without assuming specific background evolution. The rest of the variables in the UCG can be expressed in terms of $\delta\phi_{\rm UCG}$ as (we use eqs. [8]–[13]):

$$\chi_{\rm UCG} = -\frac{\gamma a^2\dot{\phi}}{2k^2H}\left[\delta\dot{\phi}_{\rm UCG} + \left(\frac{\dot{H}}{H} - \frac{\ddot{\phi}}{\dot{\phi}}\right)\delta\phi_{\rm UCG}\right], \qquad \alpha_{\rm UCG} = \frac{\gamma\dot{\phi}}{2H}\,\delta\phi_{\rm UCG}\,, \qquad \kappa_{\rm UCG} = -\frac{\gamma\dot{\phi}}{2H}\left[\delta\dot{\phi}_{\rm UCG} + \left(\frac{\dot{H}}{H} + \frac{V_{,\phi}}{\dot{\phi}}\right)\delta\phi_{\rm UCG}\right]. \tag{123}$$

The following are $\delta\phi$ in other gauges expressed in terms of $\delta\phi_{\rm UCG}$. Using a GI variable $\delta\phi_\chi \equiv \delta\phi + \dot{\phi}\chi$ (evaluate this relation in the ZSG and the UCG, which will give $\delta\phi_{\rm ZSG} = \delta\phi_{\rm UCG} + \dot{\phi}\chi_{\rm UCG}$; since the variable is GI we can evaluate it in any gauge with the value remaining the same), we can derive

$$\delta\phi_{\rm ZSG} = \delta\phi_{\rm UCG} - \frac{a^2}{k^2}\frac{\dot{H}}{H}\left[\delta\dot{\phi}_{\rm UCG} + \left(\frac{\dot{H}}{H} - \frac{\ddot{\phi}}{\dot{\phi}}\right)\delta\phi_{\rm UCG}\right]. \tag{124}$$

Using a GI variable $\delta\phi_\kappa \equiv \delta\phi + \dot{\phi}(3\dot{H} - k^2/a^2)^{-1}\kappa$, we can derive

$$\delta\phi_{\rm UEG} = \delta\phi_{\rm UCG} + \frac{\dot{\phi}}{3\dot{H} - k^2/a^2}\frac{\dot{H}}{H}\left[\delta\dot{\phi}_{\rm UCG} + \left(\frac{\dot{H}}{H} + \frac{V_{,\phi}}{\dot{\phi}}\right)\delta\phi_{\rm UCG}\right]. \tag{125}$$

Using $\delta\phi_\alpha \equiv \delta\phi + \dot{\phi}\int^t \alpha\,dt$, which has the gauge mode in it in the lower bound of the integration, we have

$$\delta\phi_{\rm SG} = \delta\phi_{\rm UCG} + \dot{\phi}\int^t \frac{\gamma\dot{\phi}}{2H}\,\delta\phi_{\rm UCG}\,dt\,, \tag{126}$$

where the lower bound of integration indicates the behavior of the gauge mode. The CG has by definition $\delta\phi_{\rm CG} \equiv 0$. Using a GI variable $\delta\phi_\epsilon \equiv \delta\phi - (\dot{\phi}/\dot{\mu})\epsilon = \delta\phi + (3H\dot{\phi})^{-1}\epsilon$, we have

$$\delta\phi_{\rm UDG} = \frac{\dot{\phi}}{3H^2}\left(\frac{H}{\dot{\phi}}\,\delta\phi_{\rm UCG}\right)^{\cdot}. \tag{127}$$

Thus, for the case of small scale we have

$$\delta\phi_{\rm ZSG} = \delta\phi_{\rm UEG} = \delta\phi_{\rm SG} = \delta\phi_{\rm UCG}\,. \tag{128}$$

The results for small scale are shown in more general context in equations (83) and (84). But, $\delta\phi_{\rm UDG}$ does not correspond to $\delta\phi_{\rm UCG}$ for the small scale. For the case of exact $H =$ constant we should refer to § 5. Similarly, we can show the inverse translations as

$$\delta\phi_{\rm UCG} = \delta\phi_{\rm ZSG} + \frac{1}{H}\left(1 + \frac{k^2}{\dot{H}a^2}\right)^{-1}\left(\delta\dot{\phi}_{\rm ZSG} - \frac{\ddot{\phi}}{\dot{\phi}}\,\delta\phi_{\rm ZSG}\right), \quad \text{etc.} \tag{129}$$

7. DISCUSSION

We saw that in the exponential and power-law inflation stages, the equation for $\delta\phi$ in the UCG simplifies to the equation based on an unperturbed metric (without the potential term). Also the perturbed scalar field equation is least complicated in the UCG, closely resembling the Minkowski space equation. It is interesting to note that the density perturbation ($\epsilon = \delta\mu$) equation in the CG has correct Newtonian correspondence, whereas, for the perturbed potential (φ) equation it is the ZSG which has correct Newtonian correspondence (see Hwang & Hyun 1994). The ϵ in the CG and φ in the ZSG are related by Poisson's equation (eq. [51]). One simple way of deriving the density perturbation spectrum generated from inflationary model is in fact by using combination of

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methods based on the UCG (for scalar field), the ZSG (for potential), and the CG (for density). We present the inflationary spectrum analysis in Hwang (1993b); based on the UCG analysis we can present the process of density fluxtuations generated from the vacuum quantum fluctuations in the inflationary era, now taking full metric fluctuations into account in analytic manner.

Let us summarize the main results presented in this paper. First, we treat perturbation analysis of a minimally coupled scalar field in a flat FLRW background for six different fundamental gauge choices. As in the case of the ideal fluid treated in H2, similar general asymptotic solutions and exact solutions for $w =$ constant cases are derived. General large- and small-scale asymptotic solutions for different gauge choices are presented in tabular form in Tables 1 and 2, respectively. The case for $w =$ constant corresponds to a power-law expanding background which can be supported by exponential potential scalar field. Exact solutions for the perturbed variables are available, and we present the exact solutions, large- and small-scale asymptotic solutions in Tables 3, 4, and 5, respectively. We identified the large-scale conservation variables available in the scalar field perturbation analysis; see § 3.8.

Second, although the whole analysis can be considered as a classical perturbation analysis, it can equally be considered as quantum field analysis in curved background, now including the simultaneously excited metric perturbations (sometimes called metric back reaction) into account. We particularly note that by including the metric perturbations the analysis become self-consistent and general treatment is available (notice that general large-scale asymptotic solutions are available in every gauge choice; see Table 1), as opposed to the case for equation (26), which is based on pure background metric. Notice that the large-scale solutions in Table 1 are valid considering general $V(\phi)$. It is important to note that neglecting the metric fluctuations, although the equation may look simpler as in equation (26), we cannot derive such asymptotic solution valid for general $V(\phi)$.

Third, in a thorough analysis for different gauges, we notice that the choice of UCG is distinguishing: the equation is most simple, and furthermore, the equation can be identified to equation (26) in the nontrivial case of power-law expanding background. In most of the gauge choices the perturbed scalar field equations do correspond to equation (26) in either the small-scale, near-exponential expansion limit, or $G \to 0$ limit. These identifications allow us easy and complete derivations of quantum fluctuations even including the full metric perturbations. Inclusion of the metric fluctuations is essential to have a consistent picture of structure generation processes based on inflation magnified vacuum quantum fluctuations. Also, this simplification of the UCG equation in a power-law expansion case provides more support on the self-consistency of including the metric perturbations: whereas the expansion stages in § 2.4 were *assumed* to be supplied by externally given source, in our case of equation (65) the simplification (eq. [122]) occurs *when* the background dynamics is supported by the background scalar field.

More detailed analysis of the curved space quantum field theory, including the metric perturbations, and the consequent clear derivation of the inflationary spectrum based on the discovery of new role of the UCG analysis in dealing with scalar field perturbations will be presented elsewhere (Hwang 1993b). Similar presentation of solutions in a class of generalized gravity theories is also possible. By adopting the UCG, similar striking simplifications will occur in the analysis of a wide class of generalized gravity theories. Previous analyses treated in Hwang (1990), H1, and Mukhanov et al. (1992) were based on the ZSG. This subject is left as a future exercise.

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